

STRATEGIES OF ENGAGEMENT
WITH SCIENTIFIC FACT-FINDING
IN INTERNATIONAL ADJUDICATION

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Amsterdam Center for International Law & University of Geneva

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Abstract:

The following contribution zeroes in on the diverging responses that permeate international adjudicative practice pertaining to international disputes arising out of scientific controversies. Drawing on the idea that scientific fact-finding is as much a struggle for argumentative persuasiveness as traditional fact-finding and law-interpretation, this article identifies and critically evaluates four attitudes of international judges and arbitrators. It shows that, when it comes to scientific fact-finding, adjudicative bodies are in a constant flux between nihilism, protectionism and outsourcing. It further demonstrates that similar dynamics can be observed with regards to the weighing of scientific knowledge in cases when adjudicators decide to outsource it to experts. This paper subsequently argues that when confronted with scientific fact-finding, international adjudicators are dealing with knowledge that is as unstable as the law and which brings them to make a choice between different types of reasoning or rationality. It ultimately makes the argument that the question of scientific fact-finding inevitably confronts international judges and arbitrators with a choice of epistemic rationality.

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Introduction

The following contribution focuses on the diverging responses that permeate international adjudicative practice pertaining to international disputes arising out of scientific controversies. Drawing on the idea that scientific fact-finding is as much a struggle for argumentative persuasiveness as traditional fact-finding and law-interpretation, this article identifies and critically evaluates four attitudes of international judges and arbitrators. It shows that, when it comes to scientific fact-finding, adjudicative bodies are in a constant flux between nihilism, protectionism and outsourcing. It further demonstrates that similar dynamics can be observed with regards to the weighing of scientific knowledge in cases when adjudicators decide to outsource it to experts. This paper ultimately argues that when confronted with scientific fact-finding, international adjudicators are dealing with knowledge that is as unstable as the law and which brings them to make a choice between different types of reasoning or rationality. It ultimately makes the argument that the question of scientific fact-finding inevitably confronts international judges and arbitrators with a choice of epistemic rationality.

This article is structured as follows. After an introductory section on the need to distinguish between law-ascertainment, fact-finding and scientific fact finding (1), a few observations will be formulated on the growing place of scientific controversy in scientific adjudication and the traditional questions of law that are potentially dependent on the prior establishment of scientific facts (2). This contribution will then turn to the identification of the four main attitudes of international judges confronted with scientific fact-finding that are witnessed in the contemporary international adjudicative practice (3). On that occasion, it will be shown that the spectrum of engagement with scientific fact-finding ranges from nihilism and scientific *non-liquet* on the one hand to outsourcing and delegation of scientific fact-finding to experts on the other. A few remarks will then be formulated on the instability of scientific knowledge and the fact that outsourcing to experts only brings about a postponement of the adjudicative responsibilities of international judges and arbitrators. It will also be demonstrated that the practice of those adjudicative bodies who defer to science and who are thus the

recipient of the scientific knowledge delivered by experts bespeaks similar oscillations between substituting legal reasoning for scientific reasoning or weighing scientific knowledge on the basis of scientific methodology (4). Drawing on the previous sections, this paper will finally argue that scientific fact-finding in international adjudication boils down to a choice between several types of epistemic rationality or reasoning. The last section will accordingly offer some critical thoughts on what it means to determine the type of rationality and reasoning process that adjudicative bodies are inevitably confronted with in cases involving questions of scientific fact-finding (5).

Before embarking on such a study, a two-fold caveat must be formulated. First, it will not come as a surprise that strategies of engagement with scientific fact-finding of domestic judges have already been the subject of literature.¹ Certainly, some of the strategies observed at the domestic level bear resemblance with those that this paper unearths and critically evaluates. However, no mechanical transposition of scholarly studies of domestic practices to the international sphere is possible because, as the following observations will show, strategies and attitudes of international judges are too specific to the international adjudicatory scene.² Although reminiscent of some of the practices pervading domestic adjudication, the considerations found here should accordingly not be generalized as they apply only to the international legal order. Second, although focused exclusively on international adjudicatory practice, this paper does not seek to provide a comprehensive overview of all cases dominated by questions of scientific fact-finding. Rather, reference is made to practice only to the extent necessary to illustrate (the politics of the) engagement strategies currently observed in international case-law. Only a limited number of decisions by international courts and tribunals are thus discussed here.

¹ See e.g. S. Jasanoff, 'Science at the Bar. Law, Science and Technology in America', Harvard University Press, 1997, esp. Chapter 2 and chapter 3. G. Edmond, 'Judging Facts : Managing Expert Knowledge in Legal Decision-Making', in G. Edmond (ed.), *Expertise in Regulation and Law*, Ashgate Publishing, 2004, pp. 136-164; F. Schauer, 'Can Bad Science Be Good Evidence: Lie Detection, Neuroscience, and the Mistaken Conflation of Legal and Scientific Norms' , 95 *Cornell Law Review* 1191 (2010).

² In the same vein, see J. Alvarez, 'Are International Judges Afraid of Science?: A Comment on Mbengue', 34 *Loyola of Los Angeles International and Comparative Law Review* (2012), at 84.

1. Legal adjudication: Law-ascertainment, fact-finding and scientific fact-finding

At this preliminary stage, it may be useful to recall that fact-finding is both intrinsic and extrinsic to any international adjudication process.³ On the one hand, it is intrinsic to the adjudicatory process since the very essence of the international judicial function is to make findings of law in light of the particular facts of a dispute. If estranged from the establishment of facts, applying the law – as it is “clarified”⁴ or “developed”⁵ by international courts and tribunals – would be a mere abstract cerebral exercise detached from all realities. On the other hand, fact-finding is extrinsic to any adjudicatory process, because the administration of international justice remains primarily contingent upon an appreciation, determination and qualification of the facts that surround a legal dispute. Indeed, without a proper reconstruction and systematization of facts, international courts and tribunals would operate in a vacuum and their decisions would address facts that are either moot or disconnected from legal disputes submitted to them. Dealing efficiently with “the complexities involved in the serious and rigorous sifting of evidence”⁶ thus requires sound fact-finding.

It is because legal adjudication is about confronting the law with facts both intrinsically and extrinsically⁷ that any adjudicator is bound to ascertain the law as well as the facts. It is a truism, however, that the techniques of establishment of the former (1.1) and those of establishment of the latter (1.2) are very divergent. Yet, the

3. See C. Chinkin, *U.N. Human Rights Council Fact-finding Missions: Lessons from Gaza*, in *LOOKING TO THE FUTURE. ESSAYS ON INTERNATIONAL LAW IN HONOR OF W. MICHAEL REISMAN* 447 (M. H. Arsanjani, J. Katz Cogan, R. D. Sloane, & S. Wiessner, eds., 2011) (“Fact-finding has often been perceived and engaged in as complementary to dispute resolution processes.”).

4. On the idea of international courts and tribunals acting as “law-clarifiers” and not as “law-makers,” see, for example, Appellate Body Report, *United States – Measures Affecting the Imports of Wool Shirts and Blouses From India*, 19 WT/DS33/AB/R (Apr. 25, 1997). See also *South West Africa Cases*, (Eth. v. S. Afr.; Liber. v. S. Afr.), Judgment, 1962 I.C.J. 465, 540 (Dec. 21) (joint dissenting opinion of Sir Percy Spender and Sir Gerald Fitzmaurice) (“It is not for a Judge today, in the light of the greater knowledge granted him by the passage of time, to do more than apply the law as it is, in the light of the facts as they stood when the situation he is dealing with arose.”).

5. See H. LAUTERPACHT, *THE DEVELOPMENT OF INTERNATIONAL LAW BY THE INTERNATIONAL COURT* xiii (1958).

6 E. LAUTERPACHT, *ASPECTS OF THE ADMINISTRATION OF INTERNATIONAL JUSTICE* 18 (1991).

⁷ For an earlier affirmation, see J. Bentham, *Rationale of Judicial Evidence : Specially Applied to English Practice : from the Manuscripts of Jeremy Bentham* (Hunt and Clarke, London 1827).

establishment of scientific facts seems to fall in neither of these two categories and follows very different modes of establishment (1.3).

1.1. The ascertainment of the law: *jura novit curia*

In most legal systems, it is commonly assumed that the judge knows the law applicable to the case submitted to its jurisdiction. This often manifests itself in the general acceptance among adjudicators and parties that the judge is supposed to know the relevant applicable rules and if not, he or she ought to unearth them: *jura novit curia*.⁸ In international law, such an idea is especially powerful when it comes to customary international rules or general principles that international judges are expected to ascertain – or unearth. Despite one century of powerful legal realist critique – a critique later buoyed by three decades of critical legal thinking – this presupposition has continued to thrive in the interpretative community of international law, thereby perpetuating the *Montesquian* myth of textualism in the judicial application of law. When it comes to the establishment of the applicable law, the judicial function has continued to be officially portrayed as limited to textual mining, that is, extracting something that is already out there. According to that view, the production of knowledge about the existing law has remained purely internal, falling in the activities on which the judge could claim a monopoly and exclusive authority.

It should be acknowledged that the principle of *jura novit curia* has been qualified in practice. Indeed, when it comes to highly technical legal issues as well as questions of law deemed of a limited geographical ambit, international courts and tribunals have been prompt to assert that the responsibility for “finding” such rules lies with the parties and no longer lies with the adjudicator, although the latter keep a

⁸ On the expression of this idea in international law, see generally J. Verhoeven, ‘Jura novit curia et le juge international’, in P.-M. Dupuy, B. Fassabender., M. Shaw, K.-P. Sommermann (eds.), *Common Values in International Law, Essays in Honour of Christian Tomuschat*, (N. P. Engel Verlag, 2006), 635-653. See for some qualification of this idea, see ICJ, *Asylum* (Colombia/Peru), 20 November 1950, ICJ Rep., pp. 226-277 and ICJ, *Rights of Nationals of the United States of America in Morocco* (France v. United States of America), 27 August 1952, ICJ Rep., 1952, p. 200. See also ICJ, *Fisheries Jurisdiction* (Federal Republic of Germany v. Iceland), 25 July 1974, ICJ Rep., 1974, p.181, ¶ 18.

supervisory function.⁹ In this situation, law-establishment comes very close to fact-finding.

1.2. The ascertainment of facts: parties-based production of knowledge under court's supervision

While one can expect the international judiciary to know the law – or if not, to extract it – it is not assumed that it knows the facts to which it will apply the law it is assumed to know or to find.¹⁰ Rather, it is seen as one of the primary functions of courts to establish such facts. In this respect, it is no longer the bringing to the surface of something already out there that is bestowed upon the international judge but the reconstruction of the reality that is supposed to have existed at the moment the dispute arose between the parties.¹¹ The law that the court is supposed to know cannot in itself produce facts and the judge is accordingly responsible for establishing them.¹² This is why it is widely assumed that before an adjudicator “can give a worthwhile legal opinion, he or she *must know the facts*”¹³ and thus get to know them. Exactly the same is true with respect to international adjudication.¹⁴

Yet, in contentious international proceedings, judges do not establish the fact themselves. They rely on the facts as evidenced by the parties to the

⁹ As early as 1950, the ICJ had highlighted that it was not supposed to know regional customs and that the establishment thereof was a responsibility of the party. See *Asylum* (Colombia/Peru), 20 November 1950, ICJ Rep, 1950, at 276-277; *Rights of Nationals of the United States of America in Morocco* (France v. United States of America), 27 August 1952, ICJ Rep., 1952, p. 200. The threshold of evidence for such regional rules seems to have been lowered over the years. See *Dispute regarding Navigational and Related Rights* (Costa Rica v. Nicaragua), 13 July 2009, ICJ Rep. 2009, para. 141; On this point, see the remarks of M. Forteau, *Les sources du droit international face au formalisme juridique*, 30 L'Observateur des Nations Unies, 2011, 61-71, at 69.

¹⁰ See C. Foster, *Science and the Precautionary Principle in International Courts and Tribunals. Expert Evidence, Burden of Proof and Finality*, Cambridge University Press, Cambridge, 2011, at 5-6. Foster explains that “the notion that it is a court or tribunal’s task to apply the law to the facts forms part of what has been characterized in the West as the ‘rationalist’ tradition”.

¹¹ *Frontier Dispute (Burk. Faso/Mali)*, (Judgment), [1986] ICJ Rep 554, ¶ 30 (Dec. 22).

¹² *Helmand River Case (Afghanistan v. Persia)*, Award of Goldsmid, Arb. (1872), in Cairo A. R. Robb ed., *International Environmental Law Reports, Volume 1, Early Decisions 8* (University of Cambridge, 1999).

¹³ S. Rosenne, ‘Fact-Finding Before the International Court of Justice’, in *Essays on International Law and Practice*, (2007), 235.

¹⁴ See gen. *Pulp Mills on the River Uruguay* (Argentina v. Uruguay), Joint Dissenting opinion I.C.J. Reports 2010.

dispute.¹⁵ In that sense, the production of the facts is generally *externalized* and *privatized*. It is a responsibility of the parties. However, such a privatization remains subject to the judge's oversight. Indeed, it is undisputed that courts and tribunals must "decide which evidence [they] choose to utilize in making [factual] findings"¹⁶ and proceed to determine "the credibility and weigh properly to be ascribed to (that is, the appreciation of) a given piece of evidence."¹⁷ In different terms, it is uncontested that there is not such a thing as "merely ascertain(ing) the facts."¹⁸ Weighing factual evidence "is part and parcel of the fact finding process."¹⁹

There have always been myriads of techniques to establish such facts.²⁰ The choice for such a technique constitutes a methodological and political choice by the adjudicative body concerned, for there is no objective fact-ascertainment. This is also why methods of establishment of facts and standards of proofs generally fluctuate in the life of all courts, especially international courts and tribunals.²¹ Such variations in the methods of fact-finding in the international adjudicatory practice have been the object of much literature and certainly do not constitute the object of the present contribution.²² It suffices to highlight here that the production of

¹⁵ See C. Foster, *Science and the Precautionary Principle in International Courts and Tribunals. Expert Evidence, Burden of Proof and Finality*, op. cit., at 80-88.

¹⁶ EC – Measures Concerning Meat and Meat Products, Appellate Body Report, WT/DS26/AB/R, WT/DS48/AB/R (Jan. 16, 1998), ¶ 135.

¹⁷ EC – Measures Concerning Meat and Meat Products, Appellate Body Report, WT/DS26/AB/R, WT/DS48/AB/R (Jan. 16, 1998), ¶ 132.

¹⁸ See the distinction drawn by the Tribunal in the *Abyei Arbitration* between the task of "merely ascertain[ing] the facts" and the task of "scientifically research[ing], select[ing] and weigh[ing] such facts" with respect to the "complex constellation of historical, anthropological and geographic facts (many of which remain obscure to this day)" that confronted the tribunal in that case. See *Abyei Arbitration (Sudan/Sudan People's Liberation Movement/Army)*, (Perm. Ct. Arb. 2009), ¶ 477, <http://www.pca-cpa.org/upload/files/Abyei%20Final%20Award.pdf> [hereinafter *Abyei Arbitration Final Award*].

¹⁹ EC Measures Concerning Meat and Meat Products, Appellate Body Report, WT/DS26/AB/R, WT/DS48/AB/R (Jan. 16, 1998), ¶ 132.

²⁰ On these techniques, see C. Foster, *Science and the Precautionary Principle in International Courts and Tribunals. Expert Evidence, Burden of Proof and Finality*, op. cit., at 88-135.

²¹ A good illustration is the change of standards of proof operated in the *Diallo* decision, which is one of the first cases in the post-Higgins era (by reference to ICJ President Rosalyn Higgins). See *Ahmadou Sadio Diallo (Republic of Guinea v. Democratic Republic of the Congo)*, 30 November 2010, ICJ Rep., 2010).

²² See A. Riddell and B. Plant, *Evidence before the International Court of Justice* (British Institute of International and Comparative Law / 2009); H. Ruiz Fabri and J.-M. Sorel (dir.), *La preuve devant les juridictions internationales* (Paris, Pedone, 2007). E. Valencia-Ospina, *Evidence before the International*

knowledge about the fact is, in international adjudicative practice, delegated to the parties under the supervision and review of the judge. The question at stake here is whether the same conclusion holds for scientific fact-finding.

1.3. Establishing scientific facts: internalizing, privatizing or externalizing?

As was explained in the previous paragraphs, the production of knowledge about the law is, subject to a few exceptions, internalized (i.e. *jura novit curia*) whilst the production of knowledge about the fact is externalized and privatized (i.e. outsourced to the parties under the court's oversight). The question of which pattern of ascertainment scientific fact-finding should follow is the central issue with which this paper is grappling.

Before recalling the lines along which scientific facts are established, it is necessary to explain why scientific fact-finding ought to be construed as a distinct category. Obviously, the “process of fact-finding”²³ is not linear. It is a circular process where international courts and tribunals always find themselves contemplating new types of facts. The nature of those facts varies substantively, and with it the nature of the process of fact-finding. In this context, one may question the existence of an autonomous category of “scientific fact-finding” in the practice of international courts and tribunals.²⁴ It is accordingly necessary to point out the specific features of scientific fact-finding.

Court of Justice?, International Law Forum du droit international; vol. 1, afl. 4, pag. 202-207 / 1999; See also M. Kamto, *Les moyens de preuve devant la Cour internationale de Justice à la lumière de quelques affaires récentes portées devant elle*, German Yearbook of International Law; vol. 49, pag. 259-292 / 2007; K. Del Mar, *Weight of Evidence Generated through Intra-Institutional Fact-Finding before the International Court of Justice*, Journal of International Dispute Settlement; vol. 2, afl. 2, pag. 393-415 / 2011; A. Aguilar Mawdsley, ‘Evidence before the International Court of Justice’, In: *Essays in honour of Wang Tieya* / Ronald St. John MacDonald gen. ed. / 1994, 533-550; M. Lachs, ‘Evidence in the procedure of the International Court of Justice : role of the Court’, In: *Essays in honour of judge Taslim Olawale Elias*, vol. 1 / 1992.

23. M. T. Grando, , *Evidence, Proof, and Fact-Finding in WTO Dispute Settlement*, Oxford University Press, Oxford, 2009, p. 242, at 5.

24. See, e.g., J. E. ALVAREZ, ‘Are International Judges Afraid of Science? A Comment on Mbengue,’ *Loyola of Los Angeles International and Comparative Law Review*, vol. 34, 2012, esp. 87-90.

A first specificity of scientific fact-finding pertains to the specific rationality required to establish scientific facts. Indeed, scientific facts classically pertain to “the structure and behavior of the physical and natural world.”²⁵ They are first and foremost substantially dependent on “scientific evidence,”²⁶ that is, “information furnishing a level of proof based on the *established and accepted methods of science*.”²⁷ They are accordingly bound to be established by recourse to scientific principles, namely “accepted fundamental laws and facts of nature *known through the methods of science*.”²⁸ Said differently, scientific fact-finding is rooted in methods of science²⁹ and not (at least not *prima facie*) in methods of law. This is precisely where the challenge posed by the establishment of scientific facts lies. Indeed, international adjudicators are not scientists,³⁰ and scientists, regardless of their potential contribution to the international dispute-settlement processes, are clearly not organs “of law”³¹ capable of or requested to settle a dispute “by the application of principles and rules of international law.”³² As international courts and tribunals are mainly composed of legal experts or arbitral practitioners³³, there is arguably *a priori* inherent limitations on the exercise of their judicial function³⁴ when they are requested to ascertain scientific facts or

25. CONCISE OXFORD ENGLISH DICTIONARY 1287 (11th ed. 2008).

26. See, e.g., BLACK’S LAW DICTIONARY 639 (9th ed. 2009) (defining “scientific evidence” as “fact or opinion evidence that purports to draw on specialized knowledge of a science or to rely on scientific principles for its evidentiary value.”).

27. See WHO, International Health Regulations art. 1.1 (2d ed. 2005) (emphasis added), http://whqlibdoc.who.int/publications/2008/9789241580410_eng.pdf [hereinafter WHO Regulations].

28. *Id.*

29. See CONCISE OXFORD ENGLISH DICTIONARY, *supra* note 35, at 1287 (defining “science” as “the intellectual and practical activity encompassing the systematic study of the structure and behavior of the physical and natural world through observation and experiment.”).

30. See, e.g., *Legality of Threat or Use of Nuclear Weapons, Advisory Opinion*, 1996 I.C.J. 237, ¶ 15 (Jul. 8) [hereinafter *Nuclear Weapons Advisory Opinion*]. The Court does not consider that, in giving an advisory opinion in the present case, it would necessarily have to write “scenarios” to study various types of nuclear weapons and to evaluate highly complex and controversial technological, strategic and scientific information. *The Court will simply address the issues arising in all their aspects by applying the legal rules relevant to the situation* (emphasis added).

31. *Nuclear Tests (Austl. v. Fr.)*, Judgment, 1974 I.C.J. 259 ¶ 23 (Dec. 20) [hereinafter *Australia Nuclear Tests Case*].

32. *Border and Transborder Armed Actions (Nicar. v. Hond.)*, Judgment, 1988 I.C.J. 89, ¶ 52 (Dec. 20).

33. *Abyei Arbitration* (Sudan/Sudan People’s Liberation Movement/Army), ¶ 482 (Perm. Ct. Arb. 2009), <http://www.pca-cpa.org/upload/files/Abyei%20Final%20Award.pdf> [hereinafter *Abyei Arbitration Final Award*].

34. *Northern Cameroons (Cameroon v. U.K.)*, Judgment, 1963 I.C.J. 29 (Dec. 2) (preliminary objections) [hereinafter *N. Cameroons case*].

data. As rightly pointed out by Arbitrator Goldsmid, “[g]eneral principles [of law] . . . are always important, but they cannot produce facts.”³⁵ This is particularly true when it comes to scientific facts.

There is another reason for distinguishing traditional law-ascertainment, fact-finding and scientific fact-finding. This relates to the specific nature of scientific fact-finding. Compared to law- and fact-establishment, scientific fact-finding must be seen as very hybrid in nature. On the one hand, it is dominantly empirical, for it pertains to the establishment of facts. It thus bears much resemblance with traditional fact-finding. On the other hand, scientific fact-finding can also be understood as a means to uncover the “non-fact” (the uncertain fact), thereby contrasting with traditional fact-finding processes before international courts and tribunals that are more orientated toward the “freezing” of facts.³⁶ In that sense, scientific fact-finding enunciates “probabilities” while traditional fact-finding methods validate “veracities”. In that sense, traditional fact-finding resembles the establishment of knowledge about the existing applicable law, more than scientific fact-finding.

The foregoing shows that scientific fact-finding features a very blended nature that distinguishes them from both law and facts: it is both an empirical and cognitive operation. One of the premises of this paper is accordingly that there exists as much similarity between scientific fact-finding and traditional fact-finding as between scientific fact-finding and law-establishment. Such a hybrid character inevitably raises the question of the mode of establishment of the scientific fact: should it follow the model of knowledge-establishment of law-ascertainment, should it be based on the model used for general fact-establishment, or should it be carried out on the basis of another method? In other words, should scientific fact-finding be left exclusively to the judge, should it be made the responsibility of the parties, or should it be outsourced to external experts? Depending on which of these methods of cognition is applied, scientific fact-finding will either resemble traditional law-establishment, come close to traditional fact-finding or grow into a wholly distinct adjudicative operation.

35. *Helmand River Case* (Afg. v. Persia), Award of Goldsmid, Arb. (1872), in *INTERNATIONAL ENVIRONMENTAL LAW REPORTS*, Volume 1, Early Decisions 8 (Cairo A. R. Robb ed., University of Cambridge, 1999).

36. *Frontier Dispute (Burk. Faso/Mali)*, (Judgment), [1986] ICJ Rep 554, ¶ 30 (Dec. 22).

Until recently, this question was mostly academic, for scientific fact-finding was most of the time incidental in cases submitted to international courts and tribunals. With the growing determinative role played by scientific fact-finding in international adjudicative practice – which is described in the following section – the question of the modes of establishment of scientific findings has become cardinal in contemporary adjudicative decision-making processes.³⁷ In addressing this question, this paper will show that, overall, the methods available for establishing scientific facts brings scientific fact-finding much closer to law-establishment than traditional fact-finding. It is important to highlight that the conclusions that will be drawn on this occasion will be equally applicable to questions of social science that can equally arise in international adjudication – for instance, in international investment arbitration – and will thus not be limited to natural sciences.

2. Ubiquity of scientific fact-finding in international adjudication

It is commonplace that the role of science and technologies is infusing all the layers of the international legal system as a whole, just as it has grown to be ubiquitous at the domestic level.³⁸ This has manifested itself in various manners. It has first and foremost impinged on the content of law, for international lawmakers have felt a need to legislate on those issues that technological and scientific developments have shown the importance of or generated themselves. As a result, international law has come to include a wide-variety of new rules on topics that had been unregulated until the advancement of science and technologies.³⁹ There even exist areas of international law – like environmental law or the law of modern communications – which owe their entire existence to the development of modern science and technology.⁴⁰ The ICJ itself has

³⁷ See C. Foster, *Science and the Precautionary Principle in International Courts and Tribunals. Expert Evidence, Burden of Proof and Finality*, *op. cit.*, at 185-239.

³⁸ S. Jasanoff, 'Science at the Bar. Law, Science and Technology in America', Harvard University Press, 1997, esp. 1-19.

³⁹ M. W. Mouton, *The Impact of Science on International Law*, (Kluwer Law International, The Hague 1999), 210-227.

⁴⁰ On this evolution, see e.g. M. Lachs, 'Views from the Bench: Thoughts on Science, Technology and World Law', 86 AJIL 673 (1992). See also J. L. Charney, 'Technology and International Negotiations', 76 AJIL 78 (1982).

acknowledged the driving force of science in international law-making when it declared “owing to new scientific insights . . . new norms and standards have been developed.”⁴¹ The opposite is true as well. Scientific disagreement can hamper law-making because of a lack of agreement as to the exact state of technological development or science.⁴² This is well-illustrated by the current impasse in international negotiations on climate change.⁴³ Eventually, scientific and technological developments have also affected the speed of international law-making processes. Indeed, they have accelerated law-making because of the sudden and universal sense of the necessity to legislate on certain issues originating in the new knowledge acquired by virtue of scientific and technological advancement.⁴⁴

It will not come as a surprise that, because more rules of international law are the result of the growing role of science and technologies in the international relations among States, the international judge has increasingly been called upon to adjudicate on their basis. This is the very reason why the question of the place of science has turned ubiquitous in international adjudication.⁴⁵ Nowadays, international adjudicators are confronted with “mass[es] of scientific and technological information,”⁴⁶ “vast amount[s] of factual and scientific material containing data and analysis,”⁴⁷ “complex scientific”⁴⁸ evidence, “highly complex and controversial technological, strategic and scientific information”⁴⁹ or again “vast mass[es] of factual material.”⁵⁰ In different

⁴¹*Gabcikovo-Nagymaros Project (Hung./Slovak.)*, Judgment, 1997 I.C.J. 7 (Sept. 25), ¶ 140.

⁴² E. Gotlieb, *Impact of Technology on Contemporary International Law*, R.C.A.D.I., Volume 170 (1981-I), 280 et seq.

⁴³ See H. Cooper and B. Knowlton, ‘Leaders delay action on climate agreement’, *International Herald Tribune*, 16 November 2009, p. 1.

⁴⁴ M. Lachs, ‘Views from the Bench: Thoughts on Science, Technology and World Law’, 86 *AJIL* 673 (1992), at 694; A. E. Gotlieb, *Impact of Technology on Contemporary International Law*, R.C.A.D.I., Volume 170 (1981-I) 128-155; M. W. Mouton, *The Impact of Science on International Law*, Kluwer Law International, The Hague 1999) 194-203.

⁴⁵ A. Riddell, ‘Scientific Evidence in the International Court of Justice – Problems and Possibilities’, 20 *Finnish Yearbook of International Law* 2009, p. 229.

⁴⁶*Gabcikovo-Nagymaros Project (Hung./Slovak.)*, Judgment, 1997 I.C.J. 7, 232, ¶ 2 (Sept. 25) (dissenting opinion of Judge Skubiszewski).

⁴⁷*Pulp Mills on the River Uruguay (Argentina v. Uruguay)*, Judgment, I.C.J. Reports 2010, ¶ 229.

⁴⁸*Pulp Mills on the River Uruguay (Argentina v. Uruguay)*, Joint Dissent, I.C.J. Reports 2010, ¶ 11.

⁴⁹*Legality of Threat or Use of Nuclear Weapons*, Advisory Opinion, 1996 I.C.J. 237, ¶ 15 (Jul. 8).

⁵⁰*Legality of Threat or Use of Nuclear Weapons*, Advisory Opinion, 1996 I.C.J. 429, 451 (July 8) (dissenting opinion of Judge Weeramantry).

terms, “[t]here is no question that modern international relations, and hence modern diplomacy and modern international litigation, is daily becoming increasingly concerned with scientific and technological facts.”⁵¹

It is noteworthy that the prominent place of scientific controversies in international disputes arising in connection with these subject-matters is not only the mere result of the central role of scientific knowledge and technological development *per se*. It also is the direct consequence of the low normativity of international legal rules designed on these questions. Indeed, in areas where science and technologies play a prominent role, many of the prescriptions and directives of conduct enshrined in the legal rules newly designed, are dependent on the existence of very contingent or variable facts: significant damage, sufficiency, reasonable threshold, necessity, reasonable utilization of shared resource, balance between economic development and environmental protection, due diligence, and any rule presupposing a risk of damage. These rules not only have a weak normative content,⁵² but also bestow additional responsibilities on international judges and call upon them to make a scientific evaluation – to be in a position to apply the applicable law to the facts. While giving way to a greater role for science, these rules of international law have simultaneously grown more inclusive of scientific uncertainty. The famous *precautionary principle* is an example of a (primary) rule providing restraint in the face of scientific uncertainty.⁵³ In contrast the principle of prevention presupposed that the risk has been objectivized.⁵⁴

⁵¹ S. Rosenne, ‘Fact-Finding Before the International Court of Justice’, in *ESSAYS ON INTERNATIONAL LAW AND PRACTICE*, 235, 237 (2007). See also Cesare Romano, ‘The Role of Experts in International Adjudication’, in *LE DROIT INTERNATIONAL FACE AUX DÉFIS ENVIRONNEMENTAUX 1* (Colloques de Aix-en-Provence 2009).

⁵² On this question see J. d’Aspremont, ‘Softness in International Law: A Self-Serving Quest for New Legal Materials’, 19 *European Journal of International Law* (2008) 1075-1093.

⁵³ *Legality of Threat or Use of Nuclear Weapons*, Advisory Opinion, 1996 I.C.J. 237, ¶ 15 (Jul. 8); *Pulp Mills on the River Uruguay (Argentina v. Uruguay)*, Judgment, I.C.J. Reports 2010. See the Separate Opinion of Judge Cancado Trindade, ¶ 62 et seq. See also Seabed Disputes Chamber of the International Tribunal for the Law of the Sea, *Responsibilities and Obligations of States sponsoring Persons and Entities with respect to Activities in the Area*, Advisory Opinion, 1 February 2011, para. 135. On the impact of the precautionary principle on the burden of proof in scientific disputes, see C. Foster, *Science and the Precautionary Principle in International Courts and Tribunals. Expert Evidence, Burden of Proof and Finality*, *op. cit.*, at 240-277.

⁵⁴ G. Hafner and I. Buffard, ‘Obligations of Prevention and the Precautionary Principle’, in J. Crawford et al (ed.), *The law of international responsibility*, (OUP 2010), 521-534; J. Vessey, ‘The

While it is uncontested that scientific fact-finding imbues international adjudication, it should be emphasized that only a few questions of law necessitate the prior establishment of scientific facts. In our view, such questions are of three different types and can be outlined as follows:

i) *The establishment of the fact at the origin of the wrongful act, the availability of exception or the establishment of circumstances precluding wrongfulness*: A good example thereof is the *Pulp Mills* case, in which scientific fact-finding was conducive to the determination that the Eastern Republic of Uruguay did not breach its substantive obligation under Article 35, 36 and 41 of the 1975 Statute of the River Uruguay. The same is true of the *Gabcikovo-Nagymaros* case.⁵⁵ Another example is the establishment of the demonstrated relevance of whale fishing by Japan within the JARPA II programme for the conservation and management of whale stocks under Article VIII of the ICRW, in the case of *Whaling in the Antarctic (Australia v. Japan)*.⁵⁶ Had the *Southern Bluefin Tuna* cases proceeded to the merits, the ITLOS would have been called upon to determine whether Japan had met its obligations to cooperate in relation to measures necessary for the conservation of the living resources of the high seas under Article 64 and 116-119 of the UNCLOS.⁵⁷ The same is true of many investment disputes in which the central question is whether a State's measures affecting trade or investment were justified.⁵⁸

ii) *The establishment of the damage actually suffered by the victim*: As an example, one can think about the case on the fumigation between

principle of prevention in international law', 3 *Austrian Review of international and european law* 181-207 (1998).

⁵⁵ *Case Concerning The Gabcikovo-Nagymaros Project, (Hungary/Slovakia)*, Judgment, (ICJ 25 September 1997).

⁵⁶ See *Whaling in the Antarctic, (Australia v. Japan)*, application instituting proceedings, Registry of the ICJ, 31 May 2010).

⁵⁷ *Southern Bluefin Tuna Cases (New Zealand v. Japan and Australia v. Japan)*, Request for Provisional Measures, Order of 27 August 1999, 117 ILR 148. See the critical remarks of C. E. Foster, 'The Consultation of Independent Experts by International Courts and Tribunals in Health and Environment Cases', 20 *Finnish Yearbook of International Law* 2009, p. 397.

⁵⁸ See *Methanex Corp. v. United States*, Final Award of the Tribunal on Jurisdiction and Merits, ¶ 102 (NAFTA Ch. 11 Arb. Trib.),

http://naftaclaims.com/Disputes/USA/Methanex/Methanex_Final_Award.pdf, Accessed 14 March 2012.

Ecuador and Colombia, wherein the establishment of the damage itself will necessitate the scientific questions be sorted out.⁵⁹

iii) The establishment of the causality between the wrongfulness and the damage: In the law of responsibility, the concept of causation has been made the measure of the primary remedial function of responsibility, i.e. reparation.⁶⁰ Indeed, the causal link between the violation and the damage calibrates the *restitutio in integrum* since the wrongdoer has the obligation to make full reparation of any damage caused by the wrongful act.⁶¹ Thus, although causality is not a constitutive element of responsibility at the international level,⁶² it plays an express role in the determination of the wrongdoing State's obligation to repair. It will not come as a surprise that the establishment of that causal link has never been reduced to a purely legal question – the reason why the ILC did not formalize it – and has often been contingent upon scientific findings. This has inevitably manifested itself in the practice of international adjudication. This was for instance the question at stake in the *Corfu Channel Case*.⁶³

⁵⁹ *Aerial Herbicide Spraying (Ecuador v. Colombia)*, application instituting proceedings (ICJ, 31 March 2008).

⁶⁰ It has been argued elsewhere that the ILC and its Special Rapporteurs nonetheless integrated another type of causality at the level of the determination of responsibility. Indeed, a specific form of causality made its way into the articles, i.e. the causal link between the conduct and the violation. On this point, see J. d'Aspremont, 'The Articles on the Responsibility of International Organizations: Magnifying the Fissures in the Law of International Responsibility', *International Organizations Law Review* (2012) (forthcoming).

⁶¹ Article 31 para. 2. See also Commentary ASR (2001), pp. 93-94. The idea that the causal link is the measure of the *restitutio in integrum* was expressly confirmed by the International Court of Justice in the *Bosnian Genocide* case, para. 462. See also the Seabed Disputes Chamber of the ITLOS in its *Advisory Opinion on the Responsibilities and Obligations of States Sponsoring Persons and Entities with Respect to Activities in the Area*, Advisory Opinion, No. 17, ITLOS, 1 February 2011, para. 181-182.

⁶² By virtue of the Anzilottian concept of wrongfulness, this type of causation was made alien to the determination of responsibility, in contrast to classical modes of determination of responsibility under domestic law. This is because the ILC and its various Special Rapporteurs deemed it an impossible task to generalize and formalize the ascertainment of the origin of every wrongful act, and accordingly decided that the determination of responsibility could not be made dependent on such a tangible concept. See the commentary of article 31 ASR, J. Crawford, *The International Law Commission's Articles on State Responsibility – Introduction, Text and Commentaries* (CUP, 2002), p. 93.

⁶³ *Corfu Channel (U.K. v. Alb.)*, Judgment, 1949 I.C.J. 18, 18 (Apr. 9).

3. Mainstream types of engagement with scientific fact-finding in international adjudicative practice

The previous sections sought to contextualize the extent to which and how questions of scientific fact-finding arise in contemporary adjudicative practice. This section attempts to show the variations in international judges' and parties' attitudes in their engagements with scientific fact-finding. It is argued here that, confronted with difficult questions of scientific fact-finding, courts and tribunals – as well as counsels – have reacted in radically different ways. Such fluctuations in the behavior of international adjudicators can be summarized by distinguishing the four types of reactions that are outlined below. These four attitudes can themselves be divided in two groups. One pertains to the general strategy of judicial *non-engagement* with scientific fact-finding (3.1). The other three constitute three variants of the overall strategy of *engagement* with scientific fact-finding (3.2).

Before embarking on such an overview, a preliminary remark is necessary. It is important to emphasize that such attitudes are not exclusive of each other. Indeed, one judicial decision can simultaneously borrow from several of them. In that sense, the engagement with scientific fact-finding can be fluctuating within the same decision. This means that one particular aspect of a decision may originate with a lesser engagement while another may be the result of a greater engagement.

3.1. The nihilist school: non-justiciability of scientific controversy (scientific non-liquet)

What we call the nihilist school is the attitude whereby a judge decides not to engage at all with scientific fact-finding because he finds that it does not behoove international courts and tribunals to deal with scientific questions in the first place. It rests on the idea that questions of scientific fact-finding are simply non-justiciable.⁶⁴ According to that view, the role of international courts and tribunals is exclusively limited to the

⁶⁴ On the question of non-justiciability, see generally O. Korhonen, 'On strategizing justiciability in international law', 10 *Finnish yearbook of international law*, (2002), 91-101; I. Brownlie, 'The justiciability of disputes and issues in international relations', 42 *British year book of international law*, (1967), 123-143.

ascertainment of the applicable rules, the establishment of the facts and the application of the former to the latter. Should the facts be unascertainable because of scientific uncertainty, the whole case becomes non-justiciable. The arguments of Japan in the *Southern Bluefin Tuna* case with respect to its admissibility are illustrative of such a strategy of non-engagement towards scientific fact-finding within international adjudication. Japan went as far as asserting that “questions of scientific judgment ... are not justiciable” and concluding that “all turn on matters of scientific, not legal, judgment.”⁶⁵ This is thus an attitude grounded in scientific *non-liquet*. Such an attitude has usually remained rather marginal in practice and, although invoked by parties, has not been espoused by international adjudicative bodies, even by those that proved reluctant to apply scientific methods of fact-establishment. Those reluctant to address scientific issues have usually, albeit under the guise of a policy of engagement, found more subtle avoidance-techniques to skirt the controversies associated with scientific fact-finding.

3.2. The Victorian engagement and its variants: prohibition of scientific non-liquet

The opposite view – which also corresponds to the mainstream approach in practice – holds that, despite the existence of “inherent limitations,”⁶⁶ international adjudicators cannot “decline to take cognizance of one aspect of a dispute merely because that dispute”⁶⁷ has scientific aspects and issues. To do otherwise would almost be tantamount to a *non-liquet* in the worst scenario.⁶⁸ International courts and tribunals are legitimately expected to act as triers of scientific facts,⁶⁹ and they can accordingly not decline to undertake such a task by asserting that

⁶⁵ *Southern Bluefin Tuna (N.Z. v. Japan, Austl. v. Japan)*, 23 R.I.A.A. 1, ¶ 40(a) (Perm. Ct. Arb. 2000).

⁶⁶ See E. LAUTERPACHT, *ASPECTS OF THE ADMINISTRATION OF INTERNATIONAL JUSTICE* (1991), at 18 (“Of course, cases can arise that make great intellectual demands upon the judges in the sense, for example, that they may require the assimilation of a mass of detailed evidence . . .”).

⁶⁷ *United States Diplomatic and Consular Staff in Tehran (U.S. v. Iran)*, Judgment, 1980 I.C.J. 3, ¶ 36 (May 24).

⁶⁸ *Abyei Arbitration (Sudan/Sudan People’s Liberation Movement/Army)*, (Perm. Ct. Arb. 2009), ¶ 479, <http://www.pca-cpa.org/upload/files/Abyei%20Final%20Award.pdf> (Accessed 14 May 2012).

⁶⁹ *European Communities – Measures concerning Meat and Meat Products (Hormones)*, Report of the Appellate Body, 19 January 1998, WT/DS26/AB/R, WT/DS48/AB/R, ¶ 133; Appellate Body Report, *EC Measures Concerning Meat and Meat Products*, WT/DS26/AB/R, WT/DS48/AB/R (Jan. 16, 1998), ¶ 132.

“disputes are scientific rather than legal.”⁷⁰ This is what we call the Victorian approach.⁷¹ The Victorian engagement to scientific fact-finding is premised on the idea that international adjudication “would be devoid of meaning if disputes concerning questions of scientific fact and opinion were not justiciable.”⁷²

The Victorian engagement with scientific fact-finding is certainly the mainstream approach to scientific fact-finding. Yet, it is important to realize that the Victorian strategy is itself the object of great variations. Indeed, there are a myriad of different ways to engage with scientific fact-finding. Three specific modes of engagement deserve to be mentioned here. We have called them the protectionist school (3.2.1), the adversarial school (3.2.2) and the outsourcing school (3.2.3).

3.2.1. The protectionist school: substituting legal argumentation for scientific argumentation (Victorian Variant 1)

What we call the protectionist school refers to the idea that international tribunals ought to be elevated to triers of scientific facts and that they ought to resolve scientific disputes according to traditional legal methods of persuasiveness. In that sense, the authority of the decision should not come from the scientific knowledge but rather from the law. The protectionist strategy is thus about preserving the arbitrary role of law and keeping scientific argumentation at bay. It is thus a strategy betraying a lack of epistemic deference⁷³ and rationality maximization.⁷⁴ It is also an approach that blurs the lines between legal methods and scientific methods, thereby manifesting some epistemic arbitrariness.

⁷⁰ *Southern Bluefin Tuna (N.Z. v. Japan, Austl. v. Japan)*, Case Nos. 2-3, Order of Aug. 27, 1999, ¶42, http://www.itlos.org/fileadmin/itlos/documents/cases/case_no_3_4/Order.27.08.99.E.pdf, Accessed 14 May 2012.

⁷¹ By reference to the Victorian school of International Law, which was embodied by Hersch Lauterpacht who advocated a prohibition of non-liquet. See generally, M. Koskenniemi, ‘Lauterpacht and the Victorian school’, 2 EJIL (1997) 215-263.

⁷² *Southern Bluefin Tuna (N.Z. v. Japan, Austl. v. Japan)*, Case Nos. 2-3, Order of Aug. 27, 1999, ¶ 41 http://www.itlos.org/fileadmin/itlos/documents/cases/case_no_3_4/Order.27.08.99.E.pdf, Accessed 14 May 2012.

⁷³ S. Brewer, ‘Scientific Expert Testimony and Intellectual Due Process’, 107 Yale Law Journal (1998) 1535-1681, at 1616.

⁷⁴ N. Luhmann, *Die Gesellschaft Der Gesellschaft* (1997), at 1088-1096.

It is well-known that this protectionist attitude has undoubtedly been the dominant feature of the International Court of Justice (hereinafter, “ICJ”)’s attitude towards scientific fact-finding. Indeed, the ICJ’s practice witnesses a very explicit move away from scientific questions. The ICJ has constantly been stifling scientific issues by hiding behind legal reasoning and using legal rationality to shield itself from scientific controversies (a). However, albeit more occasionally, such a protectionist attitude has been observed in the practice of arbitral tribunals (b).

a) The International Court of Justice (ICJ)

Scientific fact-finding has been pervading many of the contentious cases brought before the ICJ. Despite the abundance of cases where the question arises,⁷⁵ three cases are particularly symptomatic of the tendency of the Court to engage with scientific fact-finding while substituting legal reasoning for a scientific assessment of the facts of the case and which ought to be briefly recalled here.

i) Legality of the Threat or Use of Nuclear Weapons Advisory Opinion (1996): The advisory opinion of the Court on nuclear weapons is well-known and there is no need to dwell upon the findings of the Court. It suffices here to recall that the Court’s decision has been construed as the finding of a *non-liquet*, for the Court contended that international law neither prohibits nor authorizes the use of nuclear weapons.⁷⁶ That conclusion of the Court can be partly explained by the inability to establish whether *in concreto* the use of nuclear weapons would violate international law and its will to restrict itself to a finding *in abstracto*. Had the Court been willing to engage in an analysis *in concreto*, it would have been called upon to evaluate the state of the technology of nuclear weaponry as well as scientific accounts of the damage it may cause. The Court’s refusal to engage in such an inquiry is well-illustrated by the following paragraph of the opinion:

⁷⁵ Other cases are sometimes also mentioned. See, e.g., *Land and Maritime Boundary Between Cameroon and Nigeria (Cameroon v Nigeria: Equatorial Guinea Intervening)* ICJ Reports 1994; See also *Maritime Delimitation and Territorial Questions Between Qatar and Bahrain (Qatar v Bahrain)* ICJ Reports 1991; For the analysis of other science-avoidance strategies, see Anne Riddell, ‘Scientific Evidence in the International Court of Justice – Problems and Possibilities’, 20 Finnish Yearbook of International Law (2009), 229.

⁷⁶ *Legality of Threat or Use of Nuclear Weapons*, Advisory Opinion, 1996 I.C.J. 237, ¶ 105 (Jul. 8).

“The Court does not consider that, in giving an advisory opinion in the present case, it would necessarily have to write “scenarios”, to study various types of nuclear weapons and to evaluate highly complex and controversial technological, strategic and scientific information. The Court will simply address the issues arising in all their aspects by applying the legal rules relevant to the situation (emphasis added).”⁷⁷

This advisory opinion constitutes a good illustration of the protectionist attitude as the Court sought to keep scientific reasoning at bay and limit the adjudicative decision-making process as exclusively dependent on legal considerations. This is one of the reasons why the Court was severely criticized by commentators.⁷⁸

ii) *Gabcíkovo-Nagymaros Case (1997)*: In the *Gabcikovo-Nagymaros* case, a wide range of technical and scientific fields of expertise were involved, namely seismology, hydrology, hydrobiology, water chemistry, sediment transport, river morphology, soil sciences, forestry, biology, ecology and EIA methodology. The parties presented a great variety of scientific information by means of scientific studies and expert reports, as well as independent scientific appraisals by multinational construction companies involved in the dam project and by NGOs. Reports by three groups of experts established by the European Communities were also invoked. Yet, the Court decided to completely overlook this information as well as the scientific aspects of the case. Although the Court acknowledged that there was ‘abundant evidence’ that the environmental implications of the Project were ‘considerable’ and that the case entails scientific questions, it took refuge behind abstract legal reasoning. It stated: ‘it is not necessary . . . for the Court to determine which of these points of view is scientifically better founded’.⁷⁹ The Court decided to base its decision on the legal obligations of the parties arising out of treaty law as well as the circumstances precluding wrongfulness in the

⁷⁷ See, e.g., *Legality of Threat or Use of Nuclear Weapons*, Advisory Opinion, 1996 I.C.J. 237, ¶ 15 (Jul. 8).

⁷⁸ *Legality of the Threat or Use of Nuclear Weapons*, Advisory Opinion, ICJ Reports 1995, (Dissenting Opinion Weeramantry), para. 454.

⁷⁹ *Case Concerning The Gabcikovo-Nagymaros Project, (Hungary v. Slovakia)*, Judgment of 25 September 1997, para. 54.

law of state responsibility and shied away from any appraisal of scientific evidence.⁸⁰

iii) Pulp Mills Case (Argentina v. Uruguay) (2010): In this case, the Court dodged scientific controversies even more bluntly. By resorting to a distinction between procedural and substantive obligations and overlooking the extent to which they are intertwined, the Court could bypass the difficult scientific questions brought about by the case.⁸¹ Indeed, the Court indicated that it is its responsibility, after careful consideration of all the evidence placed before it by the Parties, to “determine which facts must be considered relevant, to assess their probative value, and to draw conclusions from them as appropriate.”⁸²

The criticism of that aspect of the judgment has been famously vented in the Joint Dissenting Opinion of Judges Al-Khasawneh and Simma. They have expressly bemoaned that “the Court has an unfortunate history of persisting, when faced with sophisticated scientific and technical evidence in support of the legal claims made by States before it, in resolving these issues purely through the application of its traditional legal techniques.”⁸³ According to those judges, the Court failed to use the various possibilities made available to it under the Statute and Rules.⁸⁴ Such criticisms were furthered by Judge Vinuesa.⁸⁵

The three cases mentioned above are very symptomatic of the protectionist mindset of a court that wants to restrict the dispute to an exclusive question of law. It is argued here that the ICJ's protectionism

⁸⁰ *Case Concerning The Gabčíkovo-Nagymaros Project, (Hungary v. Slovakia)*, Judgment of 25 September 1997, para. 54. The Hungarian argument on the state of necessity could not convince the Court unless it was at least proven that a real, “grave” and “imminent” “peril” existed in 1989 and that the measures taken by Hungary were the only possible response to it. Both Parties have placed on record an impressive amount of scientific material aimed at reinforcing their respective arguments. The Court has given most careful attention to this material, in which the Parties have developed their opposing views as to the ecological consequences of the Project. It concludes, however that, as will be shown below, it is not necessary in order to respond to the questions put to it in the Special Agreement for it to determine which of those points of view is scientifically better founded.

⁸¹ See the criticisms of Judge Vinuesa, dissenting opinion, *Pulp Mills on the River Uruguay (Argentina v. Uruguay)*, I.C.J. Reports 2010.

⁸² *Pulp Mills on the River Uruguay (Argentina v. Uruguay)*, Judgment, I.C.J. Reports 2010, Para. 168.

⁸³ *Pulp Mills on the River Uruguay (Argentina v. Uruguay)*, Joint Dissent, I.C.J. Reports 2010, para. 12.

⁸⁴ See contra the separate opinion of Judge Keith, *Pulp Mills on the River Uruguay (Argentina v. Uruguay)*, I.C.J. Reports 2010.

⁸⁵ *Pulp Mills on the River Uruguay (Argentina v. Uruguay)*, Judgment, I.C.J. Reports 2010, 70-71.

and tepidness to outsource scientific fact-finding does not stem from an institutional handicap. Indeed, it has been sufficiently demonstrated by legal scholars and international judges that the Statute of the ICJ as well as the Rules of Court enshrine several possibilities for outsourcing scientific fact-finding.⁸⁶ Mention has been made of Article 62⁸⁷ and 65⁸⁸ of the Rules of Court, as well as Article 50 of the Statute⁸⁹ combined with Article 67 of the Rules of Court.⁹⁰ Moreover, the case-law of the Court, as well as that of its predecessor, shows that such powers have already been used.⁹¹ The protectionist attitude of the Court described here can thus not be ascribed to institutional and procedural constraints.

It is true that the Court has occasionally resorted to so-called “shadow experts.” Sir Robert Jennings long recognized that “the Court has not infrequently employed cartographers, hydrographers, geographers,

⁸⁶ On the Court's power regarding expertise, see A. Riddel, 'Scientific Evidence in the International Court of Justice – Problems and Possibilities', 20 *Finnish Yearbook of International Law* 2009, p. 237-242. See the abovementioned opinions by Judges Simma, Vinuesa or Yusuf in the *Pulp Mills* case.

⁸⁷ Article 62 of the Rules of the Court: 1. The Court may at any time call upon the parties to produce such evidence or to give such explanations as the Court may consider to be necessary for the elucidation of any aspect of the matters in issue, or may itself seek other information for this purpose. 2. The Court may, if necessary, arrange for the attendance of a witness or expert to give evidence in the proceedings.

⁸⁸ Article 65: Witnesses and experts shall be examined by the agents, counsel or advocates of the parties under the control of the President. Questions may be put to them by the President and by the judges. Before testifying, witnesses shall remain out of court.

⁸⁹ Article 50 of the Statute: The Court may, at any time, entrust any individual, body, bureau, commission, or other organization that it may select, with the task of carrying out an enquiry or giving an expert opinion.

⁹⁰ Article 67 of the Rules: 1. If the Court considers it necessary to arrange for an enquiry or an expert opinion, it shall, after hearing the parties, issue an order to this effect, defining the subject of the enquiry or expert opinion, stating the number and mode of appointment of the persons to hold the enquiry or of the experts, and laying down the procedure to be followed. Where appropriate, the Court shall require persons appointed to carry out an enquiry, or to give an expert opinion, to make a solemn declaration. 2. Every report or record of an enquiry and every expert opinion shall be communicated to the parties, which shall be given the opportunity of commenting upon it.

⁹¹ The ICJ has mostly been resorting to Article 50. In the *Corfu Channel* case, the Court appointed a committee of three naval experts on a question of fact, contested between the parties and relevant for the question of Albania's responsibility. Once the committee had submitted its report, the Court decided that it should proceed with an *in situ* examination and submit a second report. The Court relied on the advice of a second committee in order to assess the amount of compensation owed to the United Kingdom. Moreover, in the *Gulf of Maine* case, the Chamber followed a request by the parties that it appoint a technical expert in order to assist in the delimitation of the maritime boundary. Although the appointment was made following a request by the parties, it came within the scope of Article 50. In a number of other cases, proposals by a Party to make such orders have not been accepted. The predecessor of the Court, the PCIJ, also decided at the indemnities stage of the *Chorzów Factory* case to seek expert advice before fixing the amount of compensation.

linguists, and even specialized legal experts to assist in the understanding of the issue in a case before it; and has not on the whole felt any need to make this public knowledge or even to apprise the parties.”⁹² This practice was confirmed by the Registrar of the Court, who indicated that these invisible experts are retained by the Court for purely internal consultation as that of temporary Registry staff members, entrusted with the giving of internal scientific opinions under the oath of confidentiality demanded of full-time Registry staff.⁹³ Their conclusions are never disclosed to the public. This practice has been very much criticized for its lack of transparency, openness, and procedural fairness.⁹⁴ However controversial this may be, it remains that the use of shadow experts has remained too limited a practice to offset the common avoidance techniques embraced by the Court to skirt scientific fact-finding.

b) Arbitral tribunals

The International Court of Justice is certainly not the only international adjudicative body to resort to scientific fact-finding avoidance strategy. For instance, although other aspects of that decision have displayed other attitudes towards scientific fact-finding, a similar protectionist attitude is detected in international arbitration in which, in contrast to ICJ proceedings, parties have much greater control thanks to their ability to determine the composition of the tribunal, the procedure and the applicable role. A good example of the similar practice observed in international arbitration is provided by the decision of the Arbitral Tribunal in the *Abyei Arbitration*. Indeed, the Tribunal, in this case, opted for a passive treatment of scientific fact-finding, taking into account the interpretation of its mandate as excluding “an analysis of the substantive correctness” of scientific facts.⁹⁵ The tribunal stated clearly that it would not “engage at the outset in an omnibus re-opening of the

⁹² Sir R. Y. Jennings, ‘International Lawyers and the Progressive Development of International Law’, in J. Makarczyk (ed.), *Theory of International Law at the Threshold of the 21st Century: Essays in Honour of Krzysztof Skubiszewski*, 1996, p. 416.

⁹³ P. Couvreur, ‘Le règlement juridictionnel’, in SFDI (ed.), *Le processus de délimitation maritime : Étude d’un cas fictif : Colloque international de Monaco du 27 au 29 mars 2003*, 2004, p. 349, p. 384.

⁹⁴ *Pulp Mills on the River Uruguay (Argentina v. Uruguay)*, Joint Dissenting opinion I.C.J. Reports 2010, p. 5

⁹⁵ *Abyei Arbitration Final Award*, *supra* note 9, ¶ 410.

[...] [scientific] appreciation of evidence.”⁹⁶ Also noteworthy are the consequences that the Tribunal deduced from its passive approach. While international courts and tribunals are generally reluctant *vis-à-vis* the value of scientific evidence, the *Abyei Arbitration* considered scientific fact-finding based on “scarce factual evidence,”⁹⁷ facts ascertained “regardless of the strength or weakness of the evidence uncovered,”⁹⁸ or “marked, in varying degrees, by some imprecision”⁹⁹ and by “subjective assessment”¹⁰⁰ to be highly probative. The reluctance of the Tribunal to enter into a process of scientific-finding reached its peak with the Tribunal stating evasively and without further reasoning that it “does not believe that any new evidence that has come to light is outcome-determinative.”¹⁰¹

3.2.2. The adversarial school: minimalistic engagement (Victoria Variant 2)

What we call the adversarial school refers to the approach according to which “the parties are masters of the evidence” and international adjudicators have a “passive role.”¹⁰² Although this attitude constitutes more of a textbook case, some manifestations thereof can be found in the practice of States using experts as counsels without giving them a distinct status. It is noteworthy that the Court itself has expressed concern with this practice.¹⁰³ Despite the practice of expert-counsel, such an adversarial attitude has not gained ground in international adjudicative practice. It therefore suffices to mention it without any further analysis.

⁹⁶*Id.* ¶ 411.

⁹⁷*Id.* ¶ 455.

⁹⁸*Id.* ¶ 482.

⁹⁹*Id.* ¶ 534.

¹⁰⁰*Ibid.*

¹⁰¹*Abyei Arbitration Final Award*, *supra* note 9, ¶ 538.

¹⁰²For some critical remarks on this attitude, see *Western Sahara, Advisory Opinion*, 1975 I.C.J. 127, 138 (Oct. 16) (separate opinion of Judge de Castro).

¹⁰³ *Pulp Mills on the River Uruguay (Argentina v. Uruguay)*, Judgment, I.C.J. Reports 2010, para. 168.

3.2.3. The outsourcing school: epistemic deference and the commissioning of scientific knowledge (Victorian Variant 3)

What we call the outsourcing school is the practice of international adjudicative bodies whereby they make use of scientific knowledge to reach their decisions and settle the dispute. Making use of scientific knowledge means calling upon experts to produce scientific findings geared towards the establishment of the facts necessary to solve the dispute. The outsourcing school manifests a very strong “epistemic deference”¹⁰⁴ in that judges realize that they are not familiar with the scientific questions at stake and must defer to the opinions of scientific experts.

The outsourcing school has expressed itself very vividly in the criticisms of the ICJ’s protectionist engagement with scientific fact-finding (a). However, it also seems to be a current practice (b). It is noteworthy that such an attitude has been institutionalized in the WTO dispute settlement system (c).

a) The criticisms of the protectionist approach

Among the most compelling pleas for outsourcing scientific fact-finding are those voiced through criticisms of the protective school. Indeed, criticisms of the abovementioned protective attitude have been systematically accompanied by a defense of epistemic deference in the form of solicitation of external expert opinions.

Such criticisms of the legal argumentation protectionism are not entirely new. Judge Lachs already criticized such an epistemic posture in the *North Sea Continental Shelf* case:

“Whenever law is confronted with facts of nature or technology, its solutions must rely on criteria derived from them. For law is intended to resolve problems posed by such facts and it is herein that the link between law and the realities of life is manifest. It is not legal theory which provides answers to such problems; all it does is to select and

¹⁰⁴ The expression is from S. Brewer, ‘Scientific Expert Testimony and Intellectual Due Process’, 107 Yale Law Journal (1998) 1535-1681, p. 1538.

adapt the one which best serves its purposes, and integrate it within the framework of law.”¹⁰⁵

The plea for outsourcing grew with the amplification of the unease generated by the ICJ's abovementioned protectionist attitude. It is probably on the occasion of the decision of the Court in the *Pulp Mills case* that criticisms, and thus the accompanying entreaties for outsourcing, came to a head. It suffices to recall a few of them.

For instance, Judge Yusuf declared:

“The Court, in order to exercise its function of resolving disputes, needs to ensure not only to be in possession of all the available facts relevant to the issues before it, but also to understand fully their actual meaning for the proper application of the law to those facts. The rationale behind the provisions on enquiry and the seeking of an expert opinion in the Statute and in the Rules of Court is to allow the Court to obtain the necessary assistance and support in acquiring such full knowledge of the facts.”¹⁰⁶

In the same vein, Judge Vinuesa, in his Dissenting opinion, affirmed that:

“[T]he Court reflects upon the scientific submissions by the Parties in its discussion of the data. However, throughout this overview of the evidence, there is no discussion about the scientific integrity of the scientific methodologies applied. There is also no discussion about the scientific integrity of the results. This silence on the important issue of credibility of the scientific submissions reflects more than just an accidental oversight. Instead, this silence underscores the Court's lack of scientific competence and throws doubt on the Court's ability to determine whether the data is scientifically viable or credible. The Court does not have the proper expertise or knowledge to draw the expert conclusions that it makes, and this Judgment fully reflects that.”¹⁰⁷

¹⁰⁵ See Judge Lachs' dissenting opinion in *North Sea Continental Shelf*, ICJ Rep, at 222, esp. 230. '

¹⁰⁶ *Pulp Mills on the River Uruguay (Argentina v. Uruguay)*, Dissenting Opinion of Judge Yusuf, I.C.J. Reports 2010, para. 5.

¹⁰⁷ *Pulp Mills on the River Uruguay (Argentina v. Uruguay)*, Dissenting Opinion of Judge Yusuf, I.C.J. Reports 2010, Para. 72.

In their oft-cited Joint Dissenting Opinion, Judges Al-Khasawneh and Simma dwelled on this aspect and called for a greater use of experts for the establishment of scientific facts:

“[T]he Court has clung to the habits it has traditionally followed for the assessment and evaluation of evidence to arrive at the finding in operative paragraph 2 The adjudication of disputes in which the assessment of scientific questions by experts is indispensable, as is the case here, requires an interweaving of legal process with knowledge and expertise that can only be drawn from experts properly trained to evaluate the increasingly complex nature of the facts put before the Court The Court on its own is not in a position adequately to assess and weigh complex scientific evidence of the type presented by the Parties In other words, the Court has used the traditional rules on the burden of proof and obliged Argentina to substantiate claims on issues which the Court cannot, as a court of justice, fully comprehend without recourse to expert assessment.”¹⁰⁸

They added:

“In short, in a scientific case such as the present dispute, the insights to make sound legal decisions necessarily emanate from experts consulted by the Court, even though it certainly remains for the Court to discharge the exclusively judicial functions, such as the interpretation of legal terms, the legal categorization of factual issues, and the assessment of the burden of proof.”¹⁰⁹

b) International arbitral tribunals

It is interesting to note that, subject to a few exceptions,¹¹⁰ other international bodies have not balked at addressing scientific uncertainty in the judicial process. Arbitral practice bespeaks a moderate engagement by international arbitrary tribunals with scientific fact-

¹⁰⁸ *Pulp Mills on the River Uruguay (Argentina v. Uruguay)*, Joint Dissenting Opinion, I.C.J. Reports 2010, Paras. 3-5.

¹⁰⁹ *Pulp Mills on the River Uruguay (Argentina v. Uruguay)*, Joint Dissenting Opinion, I.C.J. Reports 2010, Paras. 3-5, para. 12.

¹¹⁰ See, e.g., *Abyei Arbitration Final Award*, para note 9.

finding.¹¹¹ There are of course great variations in their engagement with scientific fact-finding, and especially in how they have weighted scientific evidence. This question is addressed below.¹¹² Whatever the varying degrees of their subsequent weighting or review of scientific evidence, there is a dominant arbitral practice to outsource scientific fact-finding to experts. It suffices here to mention a few examples.

For instance, the famous *Affaire du lac Lanoux* shows that, among international courts and tribunals, weighing scientific facts, “of which [they] have been apprized,”¹¹³ was already an accepted practice in early interstate arbitrations dealing with partially or wholly scientific questions.¹¹⁴

Yet, it is probably the Arbitral Tribunal constituted in the *Methanex* case that most recently confirmed that the ability of international judicial organs to act as weighers of facts is neither a myth nor illusory.¹¹⁵ In *Methanex*, the Tribunal opted for an active approach in its treatment of scientific facts, (i.e., it decided “to summarize the principal findings of fact which [it] has made in regard to the scientific issues relating to MTBE” (methyl tertiary-butyl ether)).¹¹⁶ It is true that one of the parties to the case expressly requested the Arbitral Tribunal to deal with the question of “whether the scientific conclusions which were presented to the Governor were so faulty that the Tribunal may reasonably infer that the science merely provided a convenient excuse for the hidden regulation of methanol producers.”¹¹⁷ The Arbitral Tribunal acted

¹¹¹ See C. Foster, *Science and the Precautionary Principle in International Courts and Tribunals. Expert Evidence, Burden of Proof and Finality*, *op. cit.*, at 123-125.

¹¹² See *infra* 4.

¹¹³ *Diversion of Water from the Meuse (Neth./Belg.)*, Judgment, 1937 P.C.I.J. (ser. A/B) No. 70, at 79 (June 28) (separate opinion of Judge Hudson).

¹¹⁴ *Affaire du lac Lanoux* (Fr./Spain), 12 R.I.A.A. 281, 367 (Trib. Arb. 1957), Lake Lanoux Arbitration (Fr./Spain), translated in 24 I.L.R. 101 (1957) ¶¶ 6 – 7. See generally C. P. R. ROMANO, *THE PEACEFUL SETTLEMENT OF INTERNATIONAL ENVIRONMENTAL DISPUTES: A PRAGMATIC APPROACH* (2000) 219–32 (analyzing the facts, judgment and aftermath of the *Affaire du lac Lanoux*).

¹¹⁵ See generally *Methanex Corp. v. United States*, Final Award of the Tribunal on Jurisdiction and Merits, ¶ **Error! Bookmark not defined.** (considering the facts of the case), (NAFTA Ch. 11 Arb. Trib.), http://naftaclaims.com/Disputes/USA/Methanex/Methanex_Final_Award.pdf, Accessed 14 March 2012.

¹¹⁶ *Id.*, ¶ 102. For a slightly different reading of the *Methanex* decision, see J. Alvarez, ‘Are International Judges Afraid of Science?: A Comment on Mbengue’, 34 *Loyola of Los Angeles International and Comparative Law Review* (2012), at 95.

¹¹⁷ *Methanex Corp. v. United States*, Final Award of the Tribunal on Jurisdiction and Merits, ¶ 19.

engaged with scientific fact-finding by ascertaining the facts and evidence before it.¹¹⁸ In a long passage—which deserves to be quoted in its entirety—the Arbitral Tribunal spelled out its engagement with scientific fact-finding:

“Having considered all the expert evidence adduced in these proceedings by both Disputing Parties, the Tribunal accepts the UC [University of California at Davis] Report as reflecting a serious, objective and scientific approach to a complex problem in California. Whilst it is possible for other scientists and researchers to disagree in good faith with certain of its methodologies, analyses and conclusions, the fact of such disagreement, even if correct, does not warrant this Tribunal in treating the UC Report as part of a political sham by California. In particular, the UC Report was subjected at the time to public hearings, testimony and peer-review; and its emergence as a serious scientific work from such an open and informed debate is the best evidence that it was not the product of a political sham engineered by California, leading subsequently to the two measures impugned by Methanex in these arbitration proceedings. Moreover, in all material respects, the Tribunal is not persuaded that the UC Report was scientifically incorrect: the Tribunal was much impressed by the scientific expert witnesses presented by the USA and tested under cross-examination by Methanex; and the Tribunal accepts without reservation these experts’ conclusions.”¹¹⁹

Another most illustrative case is provided by the *Iron Rhine Railway* arbitral decision of 24 May 2005 (Belgium v. Netherlands). In that case, the Tribunal recommended that the parties establish a committee of independent experts to determine several facts, *inter alia*, the costs of reactivating the Iron Rhine Railway, the costs of alternative autonomous development by the Netherlands, and the quantifiable benefits accruing to the Netherlands by reason of the reactivation.¹²⁰ The tribunal deemed it adequate for experts to “investigate questions of considerable scientific complexity as to which measures will be sufficient to achieve compliance with the required levels of environmental protection.”¹²¹

¹¹⁸ *Id.*, ¶ 102.

¹¹⁹ *Id.*, ¶ 101.

¹²⁰ *Award in the Arbitration regarding the Iron Rhine (“Ijzeren Rijn”) Railway between the Kingdom of Belgium and the Kingdom of the Netherlands*, Decision of 24 May 2005, VOLUME XXVII, para. 235.

¹²¹ *Id.*

Another insightful example stems from the Award of the Arbitral Tribunal of 17 September 2007 in the *Matter of an Arbitration between Guyana and Suriname*, wherein the Tribunal appointed an independent hydrographic expert and directed him as to the specific points of facts he was to examine.¹²² The Parties were given the opportunity to comment on the report of the independent hydrographic expert before the Tribunal endorsed it.¹²³

c) The WTO dispute settlement mechanism

The World Trade Organization (WTO) probably constitutes the dispute settlement mechanism in which international adjudicators have most clearly and most confidently dealt with scientific controversies.¹²⁴ A wide array of WTO panels have heard the experts put forward by the parties, have made recourse to specialized international organizations or agencies for information, or have outright heard the views of experts appointed by panels.¹²⁵ The WTO Appellate Body did not hesitate to see

¹²² Procedural Order No. 6 of the Tribunal, 27 November 2006; Order No. 7 of the Tribunal, 12 March 2007.

¹²³ Order No. 8 of the Tribunal, 21 May 2007. For some critical remarks, see A. Riddel, 'Scientific Evidence in the International Court of Justice – Problems and Possibilities', 20 *Finnish Yearbook of International Law* 2009, p. 253-255.

¹²⁴ For some general remarks, see C. Foster, *Science and the Precautionary Principle in International Courts and Tribunals. Expert Evidence, Burden of Proof and Finality*, *op. cit.*, at 114-123; C. Lévesque, 'Science in the Hands of International Investment Tribunals: a Case for "Scientific Due Process"', 20 *Finnish Yearbook of International Law* 2009, 279-283; C. E. Foster, 'The Consultation of Independent Experts by International Courts and Tribunals in Health and Environment Cases', 20 *Finnish Yearbook of International Law* 2009, p. 400-404; G. Marceau and J. K. Hawkins, 'Experts in WTO Dispute Settlement', 3 *Journal of International Dispute Settlement* (2012) (forthcoming); Y. Fukunaga, 'Standard of Review and 'Scientific Truth' in the WTO Dispute Settlement System and Investment Arbitration', 3 *Journal of International Dispute Settlement* 2012 (forthcoming).

¹²⁵ See, e.g., *European Communities – Measures Concerning Meat and Meat Products (Hormones)*, Complaint by Canada, WT/DS48/R/CAN, WT/DS26/AB/R, WT/DS48/AB/R (1998), DSR 1998:II, p. 235; *European Communities – Measures Concerning Meat and Meat Products (Hormones)*, Complaint by the United States, WT/DS26/R/USA, WT/DS26/AB/R, WT/DS48/AB/R (1998), DSR 1998:III, p. 699; *European Communities – Measures Affecting the Approval and Marketing of Biotech Products*, WT/DS291/R, WT/DS292/R, WT/DS293/R (2006) (hereinafter "EC-Biotech"); *Canada – Continued Suspension of Obligations in the EC – Hormones Dispute*, WT/DS321/R, WT/DS321/AB/R (2008); *United States – Continued Suspension of Obligations in the EC – Hormones Dispute*, WT/DS320/R, WT/DS320/AB/R (2008)). The consultation of tribunal-appointed scientific experts by WTO panels may take place even where the parties have not so requested (as in *United States – Import Prohibition of Certain Shrimp and Shrimp Products*, WT/DS58/R, WT/DS58/AB/R (1998), DSR 1998:VII, p. 2821).

itself as adequate “triers” of scientific facts.¹²⁶ It is noteworthy that the institutionalization of the engagement with scientific fact-finding within the WTO system was only partly devised by the WTO dispute settlement mechanism¹²⁷ and is mostly the product of practice.¹²⁸ Such engagement is generally a two-stage process. During the written phase, a list of questions is submitted to experts who are asked to respond individually. Parties have the possibilities to submit written comments on the experts’ replies. Questions from the panel may be submitted to the experts too. Then follows an oral consultation of experts by virtue of a joint meeting with the panel, its experts, the parties and their experts, as well as third parties.¹²⁹ All in all, the WTO dispute settlement mechanism constitutes the regime in which the outsourcing of scientific fact-finding has been the most systematically organized. Such an attitude is without doubt due to the fact that certain agreements that form an integral part of the WTO Agreement deal specifically with scientific issues. This is the case with the Agreement on Sanitary and Phytosanitary Measures (SPS) and the Agreement of Technical Barriers to Trade (TBT).

4. Scientific instability and the postponement of the reasoning process in outsourcing strategies

As was said in the previous section, there have been several types of rejection of nihilism and engagement with scientific evidence. Some have recognized the autonomy of scientific fact-finding but have still sought to substitute legal reasoning for it (*the protectionist school*). Others have left it to the parties to produce scientific evidence (*the adversarial school*). The practice, however, has witnessed a great tendency toward epistemic deference, whereas the establishment of the scientific facts is left to the expert (*the outsourcing school*). It is this third approach to scientific fact-finding with which this last section is concerned. Indeed, the epistemic deference inherent in outsourcing the production of

¹²⁶ EC – Measures Concerning Meat and Meat Products, Appellate Body Report, WT/DS26/AB/R, WT/DS48/AB/R (Jan. 16, 1998), ¶132. See also, Appellate Body Report, Korea – Taxes on Alcoholic Beverages, ¶ 161, WT/DS75/AB/R, WT/DS84/AB/R (Jan. 18, 1999) [hereinafter *Korea Alcoholic Beverages Report*].

¹²⁷ See Art 13 (2) DSU, which has been the object of limited practice. On that provision, see C. E. Foster, *Science and the Precautionary Principle in International Courts and Tribunal*, CUP, pp. 165-171 ; see also J. Pauwelyn, *The Use of Experts in the WTO*, ICLQ, 2002.

¹²⁸ C. E. Foster, ‘The Consultation of Independent Experts by International Courts and Tribunals in Health and Environment Cases’, 20 *Finnish Yearbook of International Law* 2009, p. 393.

¹²⁹ This two-stage process is usefully outlined in *ibid.*, 393-394.

knowledge on scientific fact does not come without problem. Indeed, scientific knowledge cannot be considered to be more determinate than legal knowledge (4.1). Hence, adjudicative bodies that defer scientific fact-finding to experts are inevitably confronted with very indeterminate and instable knowledge about the scientific facts at stake. The indeterminacy and instability of scientific knowledge bears the consequence that the outsourcing by adjudicative bodies only postpones their responsibilities in terms of reasoning (4.2). It will then be submitted here that epistemic deference and outsourcing come to change the nature of the contentious question that was originally brought before the commissioning adjudicative authority (4.3). Such alteration of the epistemic nature of the contention raises the question of the alteration of the reasoning process in which the arbitrator needs to engage (4.4).

4.1. The instability of scientific knowledge

It is well-known that social scientists are engaged in a competition for persuasive authority. This has contributed to the general demystification of contemporary social science, including the science of law. In law, the competition for persuasiveness and authority is a competition for the production of knowledge. The idea that the production of knowledge in social sciences is the result of a social process is no longer contested. Knowledge is even seen as being produced by power relations between professionals.¹³⁰ This is true of international law as well.¹³¹ The conclusion is not different regarding hard sciences.¹³² In that sense, even in hard sciences, production of knowledge boils down to a process aimed at securing argumentative authority among one's peers. Securing authority is parasitic to the process of communication between the actors of that community.¹³³ Indeed, as was stated above,¹³⁴ scientific fact-finding is about the enunciation of probability and scientific virtue is

¹³⁰ P. Bourdieu, 'The Force of Law: Toward a Sociology of the Juridical Field' (1987) 38 *Hastings Law Journal* 814-853, p. 827.

¹³¹ J. d'Aspremont, 'Wording in International Law', 25 *Leiden Journal of International Law* (2012) p. 575-602.

¹³² See generally J. R. Ravetz, *Scientific Knowledge and Its Social Problems*, Oxford, Clarendon Press, 1971. On the differences between legal knowledge and scientific knowledge, see S. Jasanoff, "Law's Knowledge: Science for Justice in Legal Settings", *American Journal of Public Health*, Vol. 95, No. S1, pp. S49-S58, 2005, esp. S51-S52.

¹³³ See the particular (and controversial) insights on this matter by Bruno Latour, *Science in Action* (Harvard University Press, 1987), p. 40-41.

¹³⁴ Cfr supra section 1.

inevitably a relative concept because it, as Popper taught us, holds as long as it is not falsified, refuted and rebutted.¹³⁵ Scientific facts are thus also riven by volatility, circularity, paucity, impalpability¹³⁶ and “conjectures and refutations.”¹³⁷ They are marked by disagreements “not only on the interpretation of the facts, but even on the existence or nature of at least some of them.”¹³⁸ In that sense, scientific facts are inherently unstable and indeterminate. Such instability of scientific knowledge has famously been emphasized by the WTO Appellate Body, which contended that “‘theoretical uncertainty’ . . . is inherent in the scientific method and . . . stems from the intrinsic limits of experiments, methodologies, or instruments deployed by scientists to explain a given phenomenon.”¹³⁹ In the context of a highly unstable and indeterminate scientific truth, it is not difficult to see how scientific controversy manifests argumentative struggle for persuasiveness and authority as much as legal controversies. There is not one scientific truth but only one position that needs to gain the upper hand in the struggle for persuasiveness and authority among scientists.

With respect to scientific fact-finding in international adjudication, the foregoing means that outsourcing the establishing of scientific facts to experts does not produce any monolithic knowledge. The scientific findings commissioned by the outsourcing adjudicative bodies can be extremely contested or diverging, thereby depriving that body of any firm and determinate elements on which to rely for the subsequent law-application operation necessary to adjudicate the dispute.

4.2. Postponing the reasoning process by the commissioning authority

Because of the (uncontested) instability of scientific knowledge, international adjudicative bodies can, despite some epistemic deference

¹³⁵ K. R. Popper, *Conjectures and Refutations: The Growth of Scientific Knowledge* (1963), 36.

¹³⁶ On the nature and content of scientific facts, see M. M. Mbengue, *Essai Sur Une Theorie Du Risque En Droit International Public: L'anticipation Du Risque Environnemental Et Sanitaire*, (Editions A. Pedone 2009), 97–116.

¹³⁷ Popper, *op. cit.*

¹³⁸ *Military and Paramilitary Activities in and Against Nicaragua (Nicar. v. U.S.)*, Judgment, 1986 I.C.J. 1986, (June 27), ¶ 57.

¹³⁹ Appellate Body Report, *Japan – Measures Affecting the Importation of Apples*, ¶ 241, WT/DS245/AB/R (Nov. 26, 2003) [hereinafter *Apples Appellate Body Report*].

to scientific experts, end up being the recipient of very open-ended and indeterminate findings. In such a case, there is not one single scientific fact externally ascertained that the judge can simply rubber-stamp.

Such a scientific instability and indeterminacy explains why, among those advocating for an outsourcing approach, very few believe that there is no role for the commissioning body in reviewing or weighing the scientific knowledge that has been produced for the dispute at stake.¹⁴⁰ Only a minority holds that judges should not engage in the weighing or ventilation of knowledge externally acquired.¹⁴¹ This is the *radical epistemic thesis*. According to that view, judges should merely rubber-stamp the scientific findings that have been delivered by experts. Yet, this view rests on the assumption of determinacy of scientific knowledge. If one agrees that scientific findings are necessarily as unstable as law-ascertainment and fact-determination, it can hardly be contested that the adjudicative body is bound to engage in a reasoning process about the question of which it is seized. In that sense, the epistemic deference and the outsourcing only *postpone* the reasoning of the commissioning authority. Once the commissioned and externalized fact-establishment has been completed, the adjudicative body is brought back to its adjudicative responsibilities. The epistemic deference that led the arbitrator to outsource scientific fact-finding only provided a lull in the thought-making process of the commissioning adjudicative body, which cannot escape its adjudicative responsibilities. This corresponds to a view shared by all those who are mindful of the instability and potential indeterminacy of scientific knowledge.¹⁴²

4.3. Altering the epistemic nature of the contention

As a result of scientific knowledge instability, the arbitral body which outsourced scientific fact-finding cannot skirt its adjudicative responsibilities which, during the time of outsourcing, seemed to have rest with external parties, i.e. experts. Pushed back to its adjudicative responsibilities after the completion of the outsourcing, the adjudicative body is confronted with the question of the determination of the

¹⁴⁰ See *Case concerning the Temple of Preah Vihear (Cambodia v. Thailand)*, ICJ Reports, 1962, (Separate Opinion Judge Fitzmaurice), at 58-9.

¹⁴¹ See Lachs' dissenting opinion in *North Sea Continental Shelf*, ICJ Rep, at 222, esp. 230.

¹⁴² S. Brewer, 'Scientific Expert Testimony and Intellectual Due Process', 107 Yale Law Journal (1998) 1535-1681, p. 1538.

reasoning that it ought to apply to settle the contentious question at stake. Acknowledging that outsourcing only postponed the necessity for the arbitrator to operate a settlement is not to say, however, that the externalization of the establishment of scientific fact was without effect on the whole adjudicative process. On the contrary, it is submitted here that epistemic deference and outsourcing come to change the nature of the contentious question that was originally brought before the commissioning adjudicative authority.

Whilst the nihilist school and the protectionist school mentioned above resort to strategies that do not fundamentally alter the character of the question put to the arbitrator (as the reasoning process remains of a legal nature), the outsourcing strategies recast the question in different epistemic terms. The contention is no longer as purely legal or factual as it was originally. It has become of a different epistemic nature, i.e. that of the area of knowledge to which it was outsourced. In that sense, outsourcing scientific fact-establishment inevitably transforms the nature of the dispute. In other words, the reasoning process in which the arbitrator needs to engage is not the same as the one that was brought before it. This sheds light on the fundamental difference between the nihilist and protectionist schools and the outsourcing school. Even if the latter does not elude the reasoning process and only postpones it, the epistemic deference it manifests leads to a fundamental alteration of the question put to adjudication.

4.4. Which review of scientific knowledge by adjudicative bodies?

Because the epistemic character of the question has been altered, the adjudicator must determine how to grapple with the revamped question that the experts have returned to it.¹⁴³ Indeed, the alteration of the epistemic nature of the contention raises the question of the alteration of the reasoning process in which the arbitrator needs to engage. As a result of the fundamental change of the question brought to arbitration, does the adjudicative body need to solve it by virtue of a different type of

¹⁴³ The issue of revision in scientific cases will not be dealt with in the context of the present contribution. For excellent and original developments on the topic, see C. Foster, *Science and the Precautionary Principle in International Courts and Tribunals. Expert Evidence, Burden of Proof and Finality*, *op. cit.*, at 289-303. See also *ibid.*, at 318 (discussing so-called “reassessment proceedings”).

rationality? Called upon to establish law and facts by virtue of legal rationality, does the outsourcing call on the arbitrator to apply a different rationality? This is the question of how to weigh the scientific findings produced by experts when such views are controversial or conflicting or rest on very unstable knowledge. This is the question of the determination of the rationality to be used in the phase that is called the “second-guessing.”¹⁴⁴

It will not come as a surprise that both in the literature and the case-law there is a wide variety of standards of evaluation or types of rationality that the international adjudicator ought to apply in this phase of second-guessing. In different terms, there are very diverging views as to how the commissioning adjudicative authority ought to address the unstable scientific knowledge that has been produced by the expert to which it had been outsourced. A distinction can be drawn between three types of epistemic rationality or reasoning that have been proposed to weigh scientific findings. We call them ‘intuitive rationality’ (4.4.1), ‘legal fact-ascertainment method’ (4.4.2), and ‘epistemically deferent rationality’ (4.4.3). The first one probably corresponds to the majority view while the two others have remained more marginal. A few words must be formulated about each of them. As the following observations will show, these three types of epistemic rationalities are reminiscent of the abovementioned general attitudes witnessed in connection to the engagement with scientific fact-finding as a whole. This means that the four dynamics of engagements that manifested themselves at the preliminary phase of the adjudicative process repeat themselves at the phase of second-guessing by the arbitrator.

4.4.1. Intuitive rationality

According to that view, the weighing of scientific facts – or the scientific assessment of what has happened – indisputably requires international courts and tribunals to appraise whether a “negligence,”¹⁴⁵ a “blatant

¹⁴⁴ S. Brewer, ‘Scientific Expert Testimony and Intellectual Due Process’, 107 Yale Law Journal (1998) 1535-1681, at 1616.

¹⁴⁵ This seems to be the only point on which Fitzmaurice is ready to leave some discretion to judges. See *Case concerning the Temple of Preah Vihear (Cambodia v. Thailand)*, ICJ Reports, 1962, (Separate Opinion of Judge Fitzmaurice), at 58-9.

error,”¹⁴⁶ a “fault,”¹⁴⁷ an “incorrect”¹⁴⁸ or an “unreasonable”¹⁴⁹ finding has occurred in the production of knowledge by expert to which the scientific fact-finding has been outsourced.

It is not surprising that such yardsticks for the weighing of scientific fact-finding have proved appealing to judges and observers. Indeed, they seem to refer to a rationality of great simplicity that should be in the “natural” competence of all judges and arbitrators. It is submitted here, however, that such an approach falls short of defining which exact standard of evaluation will be used to weigh and (in)validate the scientific findings. More precisely, such an approach fails to define which method of establishment of the scientific fact will be deemed “negligent,” “erroneous,” “faulty,” “incorrect” or “unreasonable.” These are all criteria which are not internally self-sufficient and which must be supplemented by an external standard of evaluation. This is why, it is argued here, that resorting to such intuitive rationality to weigh scientific fact-finding is either meant to obfuscate the political choice made by the judge or to *postpone one more time* the moment of the adjudication a bit further – that is, until another type of rationality is found and upon which the adjudicative responsibility of the bodies concerned are exercised. In that sense, it is not without a resemblance to, on the one hand, the abovementioned nihilist school and the escape from adjudicative responsibilities (the scientific *non-liquet*) and, on the other hand, the abovementioned protectionist school and its claim for exclusive adjudicative power for the arbitrator itself.

4.4.2. Legal fact-ascertainment methods

Some participants in international judicial proceedings have argued that scientific findings that are delivered to adjudicators ought to be weighed according to legal adjudicative methods of evidence – that is, on the basis

¹⁴⁶ See the rejoinder of the Respondent United States of America in *Methanex Corp. V. United States*, 23 April 2004, at para. 79.

¹⁴⁷ This seems to be the case in *id.*, para. 102.

¹⁴⁸ For a rejection of the idea that international tribunals should appraise scientific correctness, see C. Lévesque, ‘Science in the Hands of International Investment Tribunals: a Case for “Scientific Due Process”’, 20 *Finnish Yearbook of International Law* 2009, p. 260. For a similar rejection, see also *Canada – Continued Suspension of Obligations in the EC – Hormones Dispute*, para. 591 (WT/DS320/AB/R, WT/DS321/AB/R).

¹⁴⁹ This seems to be the case in the abovementioned *Methanex* arbitration, para. 102.

of the rules about the burden of proof.¹⁵⁰ According to that view, scientific fact-finding is not different from regular fact-finding and the methods of establishment of the former should follow those pertaining to the ascertainment of the latter.¹⁵¹ It leads to evaluate scientific knowledge from within the legal system. Here the resemblance with the protectionist school described above is obvious, for the arbitrator substitutes the rationality used for fact-ascertainment to the scientific rationality. In that sense, the approach described here manifests the exact same lack of epistemic deference as the protectionist school and a similar rationality maximization.¹⁵² Indeed, as is explained by Brewer, “the more a non-expert relies on his own substantive assessment of scientific evidence, the less one can be said to defer epistemically.”¹⁵³ Such a resemblance with the protectionist school shows that this way of weighing scientific findings suffers the exact same weaknesses, and especially semantic arbitrariness.¹⁵⁴

4.4.3. Radical epistemically deferent rationality

The epistemic deference at the origin of the strategy of outsourcing is mirrored in a third type of reasoning used to address scientific instability. This is the approach whereby the adjudicator relies on a type of rationality that yields entirely to that of the epistemic field from which the knowledge originates. According to that approach – which

¹⁵⁰Panel Report, *Australia - Salmon*, WT/DS18/R, ¶ 8.71, (June 12, 1998), ¶ 8.126 (emphasis added); “We cannot conduct our own risk assessment. Nor do we attempt to do so. . . . Our mandate is different. We are not asked to make a scientific risk comparison nor to state with scientific certainty that one product is riskier than the other. We can only weigh the evidence put before us and, on the basis of the rules of burden of proof we adopted, including the use of factual presumptions, decide whether sufficient evidence is before us – evidence which has not been rebutted – in order to state that it can be presumed that one product is riskier than the other.” See also *id.*, ¶ 8.41 (where the panel further stressed that “in examining this case we did not attempt (nor are we, in our view, allowed) to conduct our own risk assessment or to impose any scientific opinion on Australia. We only examined and evaluated the evidence – including the information we received from the experts advising the Panel – and arguments put before us in light of the relevant WTO provisions and, following the rules on burden of proof set out above, based our findings on this evidence and these arguments.”).

¹⁵¹ On this distinction, see the introduction above. For a defense of a similar approach in domestic systems, see F. Schauer, ‘Can Bad Science Be Good Evidence: Lie Detection, Neuroscience, and the Mistaken Conflation of Legal and Scientific Norms’, 95 *Cornell Law Review* 1191 (2010).

¹⁵² N. Luhmann, *Die Gesellschaft der Gesellschaft* (1997), at 1088-1096.

¹⁵³ S. Brewer, ‘Scientific Expert Testimony and Intellectual Due Process’, 107 *Yale Law Journal* (1998) 1535-1681, at 1616.

¹⁵⁴ See *supra* the protectionist school.

simultaneously manifests a form of evidentiary hierarchy,¹⁵⁵ the scientific findings are evaluated on the basis of the standards of the scientific community concerned. This is what could also be called the “separatist model.”¹⁵⁶ This has been the position followed by the WTO Appellate Body, which contended that its scientific findings ought to be evaluated according to methodological rigor and the standards of that scientific community¹⁵⁷ or on the basis of criteria of respectability, objectivity and coherence in the relevant scientific community.¹⁵⁸ In the same vein, the Appellate body has held that the adjudicatory body needs to determine whether a scientific fact “is sufficiently supported or reasonably warranted” by scientific evidence.¹⁵⁹

From an epistemological point of view, this approach seems the most compelling. While other standards of review of scientific fact-finding lead the commissioning adjudicative body to carry out an “epistemically arbitrary judgment” on unstable scientific findings,¹⁶⁰ this approach preserves the epistemic non-arbitrariness of the decision, for it heeds the methods whereby the knowledge is delivered by the experts and does not subject such a knowledge to an evaluation based on external standards alien to it.

That being said, this approach, although it dovetails with epistemic deference at the heart of the outsourcing school, fails to recognize that methodologies of validation of scientific knowledge are as instable as the

¹⁵⁵ On the concept of evidentiary hierarchy, see G. Edmond, ‘Judging Facts : Managing Expert Knowledge in Legal Decision-Making’, in G. Edmond (ed.), *Expertise in Regulation and Law*, Ashgate Publishing, 2004, p. 149.

¹⁵⁶ S. Jasanoff, “Law's Knowledge: Science for Justice in Legal Settings”, *American Journal of Public Health*, Vol. 95, No. S1, pp. S49-S58 (2005).

¹⁵⁷ *Canada – Continued Suspension of Obligations in the EC – Hormones Dispute*, para. 591

(WT/DS320/AB/R, WT/DS321/AB/R): “Although the scientific basis need not represent the majority view within the scientific community, it must nevertheless have the necessary scientific and methodological rigor to be considered reputable science. In other words, while the correctness of the views need not have been accepted by the broader scientific community, the views must be considered to be legitimate science according to the standards of the relevant scientific community. A panel should also assess whether the reasoning articulated on the basis of the scientific evidence is objective and coherent.”

¹⁵⁸ *Id.*

¹⁵⁹ *EC – Measures Concerning Meat and Meat Products*, Appellate Body Report, WT/DS26/AB/R, WT/DS48/AB/R (Jan. 16, 1998), ¶ 186.

¹⁶⁰ S. Brewer, ‘Scientific Expert Testimony and Intellectual Due Process’, 107 *Yale Law Journal* (1998) 1535-1681, at 1539.

knowledge itself. Indeed, validation and creation of scientific knowledge are basically the same operation. In that sense, the epistemically deferent rationality is tautological. If scientific knowledge is instable it is precisely because it cannot be “rigorously”, “coherently”, “respectably” or “objectively” validated. This approach thus rests on an illusion of scientific objective validation.

There is a second limit to this approach that pertains to what can be expected from the outsourcing adjudicative authority. Indeed, it presupposes that the adjudicative body is well-versed in the methods of the epistemic field concerned. Weighing scientific evidence on this basis depends on the epistemological competence of the adjudicative authority.¹⁶¹ In this respect, it is nonetheless hard to contest that such epistemological competence cannot be expected from judges and arbitrators at the international level, simply because they are trained for, selected and recruited on the basis of other skills.¹⁶²

5. Legal rationality v. scientific rationality and the political choice for the best argumentative persuasiveness

Section 3 above has tried to shed some light on the different attitudes witnessed in adjudicative practice with respect to judges' engagement with scientific fact-finding. It was shown the spectrum of engagement with scientific fact-finding ranges from nihilism and scientific *non-liquet* on the one hand to outsourcing and delegation of scientific fact-finding to experts on the other. Section 4 has thereafter highlighted the practice of those adjudicative bodies who defer to science and who are thus the recipient of the scientific knowledge delivered by experts. It has shed some light on the similar oscillations between substituting legal reasoning for scientific reasoning and weighing scientific knowledge on the basis of scientific methodology. The foregoing has thus demonstrated that both at the level of the determination of engagement with scientific fact-finding (section 3) and at the level of weighing the scientific evidence delivered to the tribunal (section 4), adjudicative bodies are confronted with a choice of reasoning, that is a choice of the rationality by virtue of which (and under the guise of) their decision will be made. This last

¹⁶¹ *Ibid.* at 1681.

¹⁶² See gen. D. Terris, C. P. R. Romano and L. Swigart, *The International Judge, An Introduction to the Men and Women Who Decide the World's Cases* (OUP, 2007).

section aims to offer some critical thoughts on this choice of rationality and reasoning process that adjudicative bodies face in cases involving questions of scientific fact-finding.

With a view to reflecting upon the choice of reasoning process that adjudicative bodies inevitably need to make to address questions of scientific fact-finding, it is necessary to indicate that, in our view, international tribunals are primarily administrative mechanisms to settle international disputes on the basis of an exercise of authority. More particularly, their machinery is geared towards the application of a given sort of rationality or reasoning process – i.e. legal rationality – to a dispute between two parties to which both expect this administrative machinery to produce an authoritative solution.¹⁶³ And it is only if the decision is perceived as authoritative that parties will abide by it and continue to yield in the future to the dispute settlement mechanism concerned.

It is submitted here that the efficacy and authority of the administrative machinery of international courts and tribunals – and that of the judicial answer that it provides – hinges on the extent to which they (and the rationality they apply) are perceived as legitimate. This is certainly truistic for international lawyers. In an international legal system where there is in general no compulsory jurisdiction, these courts and tribunals, if they are not perceived as legitimate or applying a legitimate rationality, are quickly left without clients. More specifically, international actors that are eligible for international dispute settlement mechanisms will submit cases involving scientific aspects before international courts only to the extent that they are confident that their case will be fully and duly appreciated by judges.¹⁶⁴ According to this approach to dispute settlement, it can be said that international adjudicative bodies are vying for argumentative authority. They must constantly nurture and sustain the authority of their decisions. Again, this is certainly not unheard of. This elementary digression is only meant to make the following point. Against the backdrop of such an –

¹⁶³ For similar views on domestic adjudicative bodies, see S. Jasanoff, 'Science at the Bar. Law, Science and Technology in America', Harvard University Press, 1997, p. 19.

¹⁶⁴ A. Riddel, 'Scientific Evidence in the International Court of Justice – Problems and Possibilities', 20 *Finnish Yearbook of International Law* 2009, p. 252. In the same vein, see C. E. Foster, 'The Consultation of Independent Experts by International Courts and Tribunals in Health and Environment Cases', 20 *Finnish Yearbook of International Law* 2009, p. 404.

uncontested – quest for authority on the part of international adjudicative bodies, the question of scientific fact-finding becomes a question of determination as to *which type of rationality is the most likely to generate persuasiveness of the decision* and hence legitimacy and authority and efficacy of international dispute-settlement bodies. In other words, in the search for argumentative authority, which is the type of reasoning process that is the most conducive to the persuasiveness, legitimacy, authority and efficacy of international adjudication?

It is true that deference to the epistemic field from which the knowledge is deemed to emanate should, from an epistemic point of view, be preferred. However, it is argued here that, in the context of international adjudication, none of the various strategies of engagement with scientific fact-finding and none of the standards of evaluation of scientific knowledge necessarily trump the other. In our view, the determination of engagement with scientific fact-finding and the weighing thereof is all contingent upon various parameters, including the functions and types of adjudication, the powers of the adjudicative body and the circumstances and facts of the case. There is thus no hierarchy of reasoning processes *in abstracto*. It is all a choice that must be made *in concreto* on a case-by-case basis.

Even though it remains agnostic as to the choice of rationality to be applied to engage with and weigh scientific fact-finding, the argument made here nonetheless takes issue with attempts to blur the lines between the various epistemic rationalities to which the adjudicative may be resorting. Indeed, it is submitted here that obfuscating the lines between the epistemic rationalities – a result of the abovementioned protectionist school or the use of legal methods of evidence for the weighing of scientific fact-finding – brings about epistemic arbitrariness. Epistemic non-arbitrariness requires that a strict separation between scientific and legal rationalities be maintained.¹⁶⁵ Yet epistemic non-arbitrariness is, in our view, the very prerequisite of legitimacy of the adjudicative decision-making process.¹⁶⁶ It is only to the extent that (potential) parties can perceive the rationality to which the adjudicative

¹⁶⁵ Some people have argued that this was not possible. See C. E. Foster, 'The Consultation of Independent Experts by International Courts and Tribunals in Health and Environment Cases', 20 *Finnish Yearbook of International Law* 2009, p. 391-392.

¹⁶⁶ S. Brewer, 'Scientific Expert Testimony and Intellectual Due Process', 107 *Yale Law Journal* (1998) 1535-1681, at 1680.

body resorts and simultaneously believes that this body is sufficiently competent to apply such reasoning that they will abide by its decision and continue to make use of the dispute settlement mechanism concerned. The question at stake in cases of scientific fact-finding is thus all about ensuring *epistemic non-arbitrariness*.

This paper is certainly not the place to formulate recommendations for guaranteeing epistemic non-arbitrariness in international adjudicative practice.¹⁶⁷ There are a great variety of ways in which epistemic non-arbitrariness can be pursued. However, among all the epistemic non-arbitrariness-enhancing mechanisms that have been mentioned in the literature, two deserve some attention here.

First, it is interesting to recall that epistemic non-arbitrariness has been seen as dependent upon an “intellectual due process”¹⁶⁸ – that is, an argumentative methodology that allows a strict separation between the different epistemic analyses to which the arbitrators could be resorting.¹⁶⁹ This is thus an invitation for an argumentative formalism that allows the recipient of the judgment to identify – and hence evaluate – which epistemic rationalities have informed the arbitrator’s adjudicative decision-making process. Such an intellectual due process is thus a type of formalism that makes the legal argumentation of international arbitrators more transparent and universally accessible.¹⁷⁰

¹⁶⁷ Foster’s visionary work includes recommendations for ensuring epistemic non-arbitrariness in international adjudicative practice. See, in particular, C. Foster, *Science and the Precautionary Principle in International Courts and Tribunals. Expert Evidence, Burden of Proof and Finality*, *op. cit.*, at 240-277.

¹⁶⁸ S. Brewer, ‘Scientific Expert Testimony and Intellectual Due Process’, 107 *Yale Law Journal* (1998) 1535-1681, at 1539.

¹⁶⁹ *Ibid.* at 1539.

¹⁷⁰ In that sense, it bears a lot of resemblance to Martti Koskenniemi’s famous concept of “culture of formalism.” M. Koskenniemi, *The Gentle Civilizer of Nations: The Rise and Fall of International Law 1870-1960* (Cambridge University Press, 2002), at 500-508. For a discussion of that concept, see E. Jouannet, ‘Présentation critique’, in M. Koskenniemi, *La Politique du Droit International* (Paris, Pedone, 2007), at 32-33. See also I. de la Rasilla del Moral, ‘Martti Koskenniemi and The Spirit of the Beehive in International Law’, 10 *Global Jurist* (2010); J. von Bernstorff, ‘Sisyphus was an international lawyer. On Martti Koskenniemi’s “From Apologia to Utopia” and the place of law in international politics’, 7 *German Law Journal* (2006) 1015, at 1029-1031; J. A. Beckett, ‘“Rebel Without a Cause” Martti Koskenniemi and the Critical Legal Project’, 7 *German Law Review* (2006) 1045; see also the book review of Martti Koskenniemi, *The Gentle Civilizer of Nations: The Rise and Fall of International Law 1870-1960* by N. Tsagourias, 16 *Leiden Journal of International Law* (2003) 397, pp. 398-399. For some critical remarks on the concept of culture of formalism, see J. d’Aspremont, *Formalism and the Sources of International Law* (OUP, 2011), 27-29.

Another epistemic non-arbitrariness-enhancing mechanism that has been more commonly referenced in the literature originates in the belief that procedural guarantees of transparency, could suffice to create a use of experts that is open and transparent, and ensures that scientific evidence has not been arbitrary. Under the name of “scientific due process,”¹⁷¹ some scholars have thus been making a plea for procedural rules for the use of experts with a view to ensuring openness and transparency. Such an approach is supposedly meant to allow a tribunal to apply scientific knowledge produced by experts without having to decide whether the scientific conclusions were correct.¹⁷² This is simultaneously supposed to allow legal rationality and scientific rationality to be clearly delineated and not mixed.¹⁷³

It cannot be excluded that these two particular epistemic non-arbitrariness-enhancing mechanisms prove conducive to alleviating epistemic arbitrariness in cases involving scientific fact-finding. It is argued here that the methodological approach based on the idea of intellectual due process is certainly the one that could be most efficient in this respect. Indeed, it is probably the only approach that could ensure genuine transparency in decisions of adjudicative bodies in connection with scientific fact-finding. Moreover, it is not certain that procedural guarantees alone will suffice to ensure epistemic non-arbitrariness. This being said, such an intellectual due process seems to be at best a horizon. The myth of full transparency of legal argumentation – which is at the heart of some critical strands of the international legal scholarship – is bound to remain an ideal. This is why we believe that the authority and legitimacy of adjudicative decision-making processes involving questions of science is accordingly condemned, in the light of the argument made here, to be in constant fluctuation. Leaving it to the adjudicative body to astutely choose the rationality that is the most instrumental to

¹⁷¹ C. Lévesque, ‘Science in the Hands of International Investment Tribunals: a Case for “Scientific Due Process”’, 20 *Finnish Yearbook of International Law* 2009, p. 260. For some similar views, see C. E. Foster, ‘The Consultation of Independent Experts by International Courts and Tribunals in Health and Environment Cases’, 20 *Finnish Yearbook of International Law* 2009, p. 404-405.

¹⁷² C. Lévesque, ‘Science in the Hands of International Investment Tribunals: a Case for “Scientific Due Process”’, 20 *Finnish Yearbook of International Law* 2009, p. 287.

¹⁷³ *Ibid.*, p. 290. See contra C. E. Foster, ‘The Consultation of Independent Experts by International Courts and Tribunals in Health and Environment Cases’, 20 *Finnish Yearbook of International Law* 2009, p. 391.

preserving a perception of epistemic non-arbitrariness in the case before it.

6. Concluding remarks

As the foregoing shows, there is no silver bullet to solve the clash of rationalities inherent in cases involving scientific fact-finding with a significant scientific dimension. If it is willing to not elude its adjudicative responsibilities through a nihilist approach,¹⁷⁴ an adjudicative body is inevitably bound to determine which rationality or reasoning process it will wield to reach its decision. That choice, it has been argued here, cannot be made *in abstracto*. This contribution has thus remained agnostic as to the determination of the “best” rationality and reasoning process in contentious cases involving scientific questions. That choice, in our view, hinges on the functions and types of adjudication, the powers of the adjudicative body and the circumstances and facts of the case. It has, however, been argued here that in making that choice, adjudicative bodies should do their utmost to aim for epistemic non-arbitrariness, since it is a prerequisite for the legitimacy and authority of the adjudicative decision-making process.¹⁷⁵ Indeed, it is only to the extent that parties can perceive the epistemic rationality to which the adjudicative body resorts and simultaneously believe that this body is sufficiently competent for applying such a reasoning process, that their trust and confidence in the dispute settlement mechanism concerned will endure. In different terms, in making the choice for a given reasoning process in cases involving scientific fact-finding, international courts and tribunals should remain very clear about the language they speak and the rationality they apply to settle the disputes submitted to them. Indeed, it is probably one of the greatest lessons learnt from legal realism at the beginning of the 20th century that there is nothing more detrimental to the legitimacy, authority and efficacy of dispute settlement than an adjudicative body deceiving the justices as to the language it speaks. The omnipresence of scientific fact-finding in international adjudication constitutes a useful opportunity to recall the continued relevance of this elementary consideration by the legal realist school a century ago.

¹⁷⁴ Cfr. *supra* 3.1.

¹⁷⁵ S. Brewer, ‘Scientific Expert Testimony and Intellectual Due Process’, 107 *Yale Law Journal* (1998) 1535-1681, at 1680.